

Thomason Model Structures in Every Strict Higher Dimension

OpenAI

September 25, 2026

Abstract

We prove the higher-dimensional Thomason model-structure conjecture of Ara and Maltinis. For every $1 \leq n \leq \infty$, the category of small strict globular n -categories admits a proper combinatorial model structure that is Quillen equivalent to simplicial sets. Its weak equivalences and fibrations are detected by the twice-extended Street nerve $\mathrm{Ex}^2 N_n$, and the Quillen equivalence is given by $c_n \mathrm{Sd}^2 \dashv \mathrm{Ex}^2 N_n$. Thus strict higher categories model the homotopy theory of spaces in every positive finite dimension and in dimension ω .

Contents

1	Introduction	1
2	Strict categories, directed chains, and simplicial gluing	4
2.1	The directed tensor convention	5
2.2	Right cylinders and truncation	6
2.3	The Alexander–Whitney map	7
2.4	Finiteness and gluing in simplicial sets	8
3	Positive relative prisms	10
4	Labelled pieces of algebraic pushouts	14
5	Couniversality and the unit on posets	17
6	Comparison for subdivided attachments	21
7	The model structure and the Quillen equivalence	24
A	Presentations and local finite presentability	28
A.1	Operations and their domains	29
A.2	Defined terms and presentations	30
A.3	Limits, colimits, and finite presentations	31
B	Relation to the weak-groupoid expansion method	32

1 Introduction

Thomason’s model structure equips the category of small categories with the homotopy theory of spaces. Its higher-dimensional analogue asks for the same result on the entire category of strict

globular n -categories, with weak equivalences detected by the Street nerve and fibrations detected by its twice-extended nerve. We prove this analogue in every dimension, resolving positively the higher-dimensional Thomason model-structure conjecture of Ara and Maltsiniotis.

Fix a universe of small sets and enlarge the universe when forming categories of small objects. For a positive integer n , let $n\text{Cat}$ be the category of small strict globular n -categories and strict functors. We regard it as the full subcategory of ωCat consisting of strict ω -categories whose cells above dimension n are identities; for $n = \infty$, set $n\text{Cat} = \omega\text{Cat}$. There is no invertibility assumption on positive-dimensional cells. The Street nerve and its left adjoint are denoted by

$$c_n : \text{sSet} \rightleftarrows n\text{Cat} : N_n, \quad (N_n X)_m = \text{Hom}_{\omega\text{Cat}}(\mathcal{O}_m, X),$$

where $\mathcal{O}_m$ is the m th oriental. With the usual subdivision adjunction $\text{Sd} \dashv \text{Ex}$, put

$$L_n = c_n \text{Sd}^2, \quad R_n = \text{Ex}^2 N_n.$$

Write W_{KQ} and Fib_{KQ} for the weak homotopy equivalences and Kan fibrations of simplicial sets, and define

$$W_n = R_n^{-1}(W_{\text{KQ}}) = N_n^{-1}(W_{\text{KQ}}), \quad F_n = R_n^{-1}(\text{Fib}_{\text{KQ}}), \quad C_n = {}^\perp(F_n \cap W_n). \quad (1.1)$$

Here ${}^\perp$ denotes the left lifting class. The equality defining W_n follows from the classical properties of Ex , recalled in Section 7.

Theorem 1.1. *For every $n \in \{1, 2, 3, \dots\} \cup \{\infty\}$, the classes (C_n, W_n, F_n) in (1.1) form a proper combinatorial model structure on $n\text{Cat}$. It is cofibrantly generated by $(L_n I, L_n J)$, where*

$$I = \{\partial \Delta[m] \hookrightarrow \Delta[m] : m \geq 0\}, \quad J = \{\Lambda^k[m] \hookrightarrow \Delta[m] : m \geq 1, 0 \leq k \leq m\}.$$

Moreover,

$$L_n : \text{sSet}_{\text{KQ}} \rightleftarrows (n\text{Cat}, C_n, W_n, F_n) : R_n$$

is a Quillen equivalence.

History and significance

Thomason established the case $n = 1$ by transferring the simplicial model structure through $c_1 \text{Sd}^2 \dashv \text{Ex}^2 N_1$ [17]. His proof separates two issues that persist in higher dimensions: the homotopy behavior of the adjunction unit and the comparison between a pushout of nerves and the nerve of a categorical pushout. For the latter he introduced Dwyer maps and proved the comparison for their arbitrary cobase changes [17, Definition 4.1 and Proposition 4.3]. Cisinski corrected the retract argument used for properness, while preserving the model structure and its properness [7]. Street's orientals extend the nerve to strict higher categories [16], providing the weak equivalences in the question considered here.

Ara and Maltsiniotis developed an abstract transfer theorem, drawing on the homotopy theory of categories developed by Grothendieck and Cisinski, and established the proper combinatorial model structure in dimension two. Using Chiche's comparison theorem for the homotopy categories of categories and 2-categories, they also obtained the Quillen equivalence [4, Theorems 4.11, 6.27, 6.30, and Corollary 6.32]. Worytkiewicz, Hess, Parent, and Tonks had proposed the two-dimensional structure earlier [18]; their later corrigendum addresses the gaps identified by Ara and Maltsiniotis [19]. For arbitrary $1 \leq n \leq \infty$, the transfer framework isolates two sufficient conditions: a unit

comparison on nerves of posets, and preservation of certain poset equivalences by every categorical pushout [4, Scholie 5.14].

Ara and Maltsiniotis proved the poset-unit condition in all these dimensions [5, Corollary 10.11 and Scholie 10.12]. Gagna subsequently gave a direct simplicial proof and showed that the Street nerve induces an equivalence with homotopy types after localizing at the nerve weak equivalences [9, Theorems 5.6 and 6.9]. Thus strict higher categories already described the homotopy category of spaces in every dimension. The prescribed model structure requires the additional lifting and factorization statements, whose construction depends on controlling actual categorical pushouts. Ara’s 2023 account records this remaining sufficient condition and the general conjecture [3, §§9.2.5–9.2.6]; Guetta and Maltsiniotis also state the ω -dimensional model-structure conjecture explicitly [11, Introduction, p. 6].

The shapes of higher cells matter in this question. Fiore and Paoli constructed a Thomason model structure on small n -fold categories, whose weak equivalences are detected by the diagonal of the n -fold nerve and which is Quillen equivalent to simplicial sets [8, Theorem 9.28]. Their objects are obtained by iterating the internal-category construction; the present theorem concerns strict globular categories and the Street nerve.

Theorem 1.1 supplies both weak factorization systems on all strict n -categories, together with left and right properness, by establishing the remaining pushout condition. Its arbitrary-target scope is essential: the small object argument must allow attachments to categories that need not be cofibrant or freely generated. The ω -dimensional proof uses finite orientals directly, and finite-dimensional reflection carries the same construction to each finite n .

Stationary cylinders and the pushout argument

For a poset E , write $U_E = c_n N E$, where N is the ordinary nerve. The principal intermediate result is Theorem 5.2: if $A \subseteq E$ is a downward-closed full subposet and its inclusion has a right adjoint retraction, then every pushout of $U_A \rightarrow U_E$ is a Street weak equivalence. This proves the sufficient condition isolated in [4, Scholie 5.14, condition (d’)]. The cylinders used for contraction fix U_A throughout the interval. After any attachment $U_A \rightarrow X$, this equality lets each cylinder glue to the constant cylinder on X , controlling the entire pushout and its newly formed composites.

The construction starts with a restricted but explicit directed prism. Suppose $f, g : E \rightarrow E$ are monotone, $f \leq g$, they agree on a sieve A , and g is constant on $E \setminus A$. Write u for the degree-one generator of the interval chain complex. For every increasing tuple w , the formula

$$[w_0, \dots, w_p] \otimes u \mapsto [fw_0, \dots, fw_p, gw_p]$$

together with the endpoint maps induced by f and g defines a positive chain map after normalized repetitions are set to zero. It produces a strict Gray cylinder constant in the interval direction on U_A , at the common endpoint restriction induced by $f|_A = g|_A$. Its positivity is a consequence of the precise hypotheses on the old part and the constant upper map. Section 3 proves the chain identity, compatibility with all simplex operators, and descent to every finite dimension.

When $E \setminus A$ has a greatest element, we choose two such cylinders whose defining poset maps restrict to id_A . They give a relative contraction by a zigzag, which extends across any attachment. For a general finite poset, deleting distinct maximal new elements gives two sieves containing A . The corresponding pushouts are full sieve subcategories of the total attachment, with intersection the pushout for the common subposet. Their nerves form an ordinary pushout along monomorphisms, so the local weak-equivalence statements glue. This allows each local contraction to be chosen independently. Finite subposets stable under the retraction then yield the result for arbitrary posets.

The remaining comparison follows the geometry of Thomason’s Dwyer maps [17, Propositions 4.2–4.3]. Two subdivisions put each generating boundary or horn inclusion into the form $NA \hookrightarrow NB$, where A is a sieve and its inclusion into its upward closure in B has a right adjoint retraction (Lemma 6.2). For any small poset pair with these properties, Proposition 6.1 provides a reusable comparison between the pushout of Street nerves and the nerve of an attachment along an arbitrary functor $U_A \rightarrow X$. The poset description here concerns the generating pieces.

At the chain level we use Steiner’s description of orientals [14, 15] and the biclosed Gray tensor product, with the completed extension theorem of Ara and Maltsiniotis [6, Theorem A.15]. Positive-chain constructions already give simplicial nerve homotopies in Gagna’s proof [9, Sections 6.3–6.8]. Here the restricted prism hypotheses give strict relative cylinders; the choices fixing A extend across arbitrary attachments. The cover calculations build on the fiber and sieve machinery of Ara and Maltsiniotis [4, Lemma 2.8 and Propositions 5.3 and 5.5]; they allow us to glue the weak-equivalence conclusions without choosing compatible contractions on overlapping pieces.

Organization

Section 2 records the foundations and the precise simplicial gluing lemma. Sections 3 and 4 construct the stationary cylinders and identify labelled pieces. Section 5 proves couniversality and the unit comparison for posets. Section 6 proves the nerve comparison for subdivided boundary and horn attachments, with explicit control of the monomorphic gluing legs. Section 7 supplies the model axioms, properness, and the unit comparison for arbitrary simplicial sets. Appendix A proves the required presentability and smallness assertions by guarded term presentations. Appendix B discusses the relationship with the elementary-expansion method for Grothendieck groupoids; its claims are not inputs to the proof.

2 Strict categories, directed chains, and simplicial gluing

The pushout argument needs three foundations: directed chain maps must give strict right cylinders, these cylinders must descend to finite dimensions, and their nerve comparisons must survive gluing and filtered colimits. We fix the conventions and prove the needed local statements before constructing the relative prism.

We work with small objects in a fixed universe. An ω -category in this paper is a strict globular ω -category: its cells have globular sources and targets, identities, and compositions along matching lower-dimensional boundaries, satisfying the associativity, unit, and interchange laws strictly. An n -category, for finite n , has only identity cells above dimension n . The inclusion $n\text{Cat} \hookrightarrow \omega\text{Cat}$ has a left adjoint τ_n . Explicitly, $\tau_n X$ retains the cells of X in dimensions below n , and its n -cells are the n -cells of X modulo the congruence generated by

$$s_n(z) = t_n(z) \quad (z \text{ an } (n+1)\text{-cell of } X).$$

Here the congruence is closed under every defined composition. Its generating pairs are parallel n -cells, so their identifications leave every lower matching boundary unchanged; compositions therefore descend to the quotient. Cells above n become identities. Every strict functor from X to an n -category annihilates these relations, which proves the stated reflection property. We set $\tau_\infty = \text{id}$ and use $n\text{Cat} = \omega\text{Cat}$ when $n = \infty$.

Write D_i for the free strict ω -category on one i -cell, with all its globular faces. Thus $\text{Hom}_{\omega\text{Cat}}(D_i, X)$ is the set of i -cells of X , D_0 is terminal, and D_1 is the category $0 \rightarrow 1$. Let $\mathcal{O}_m$ denote the m th oriental. Its standard cosimplicial structure defines

$$(N_n X)_m = \text{Hom}_{\omega\text{Cat}}(\mathcal{O}_m, X) = \text{Hom}_{n\text{Cat}}(\tau_n \mathcal{O}_m, X).$$

For ω -categories we also write N_∞ . The left adjoints are given by the simplex colimits

$$c_\infty K = \operatorname{colim}_{(\Delta[m] \rightarrow K) \in \Delta/K} \mathcal{O}_m, \quad c_n K = \tau_n c_\infty K = \operatorname{colim}_{(\Delta[m] \rightarrow K) \in \Delta/K} \tau_n \mathcal{O}_m. \quad (2.1)$$

These formulas follow from the universal property of a colimit and the degreewise definition of the nerve. In particular, for a poset E ,

$$U_E^\infty = c_\infty N E, \quad U_E = c_n N E = \tau_n U_E^\infty.$$

The fixed dimension is suppressed in U : for a monotone map f , write $U(f) = c_n N(f)$ and $U^\infty(f) = c_\infty N(f)$. We use N for the ordinary nerve. On ordinary categories it agrees with the Street nerve: a map from an ordinal is determined by its successive edges, because its triangular cells impose the usual composition equations. In particular $N_\infty D_1 = \Delta[1]$.

2.1 The directed tensor convention

An augmented directed complex is a nonnegative chain complex K of abelian groups, with augmentation $\epsilon : K_0 \rightarrow \mathbb{Z}$ and a specified additive submonoid $K_p^+ \subset K_p$ in each degree. Its morphisms are chain maps preserving the augmentation and these submonoids. All complexes used below have a specified free basis, with positive submonoids generated by the basis. For such a complex write $dc = d^+c - d^-c$ with disjoint positive supports. A basis is *unital* when $\epsilon((d^-)^p b) = \epsilon((d^+)^p b) = 1$ for each basis element b of degree p , and is *strongly loop-free* when the relations

$$a < b \quad \text{if } a \text{ occurs in } d^-b, \quad b < a \quad \text{if } a \text{ occurs in } d^+b$$

are contained in a strict partial order on the whole basis. A complex with such a basis is called a *strong Steiner complex*.

Steiner's functor ν sends augmented directed complexes to strict ω -categories. Its cells retain the lower-dimensional boundary data of a chain. More concretely, a p -cell of νK is a table

$$(x_0^-, x_0^+ \mid x_1^-, x_1^+ \mid \cdots \mid x_p^-, x_p^+), \quad x_r^\pm \in K_r^\pm, \quad (2.2)$$

such that

$$\epsilon x_0^- = \epsilon x_0^+ = 1, \quad dx_r^- = dx_r^+ = x_{r-1}^+ - x_{r-1}^- \quad (1 \leq r \leq p), \quad x_p^- = x_p^+.$$

An identity appends a zero pair to the table. A positive, augmentation-preserving chain map acts on every entry, giving the corresponding strict functor [14, Definitions 2.6 and 2.8, Proposition 2.7]. For a basis element b of degree p , its *atom* has top pair (b, b) and entries $\langle b \rangle_r^\pm = (d^\pm)^{p-r} b$ for $0 \leq r \leq p$: at each step, take the negative or positive part of the boundary. For a strong Steiner complex these tables are cells, and νK is freely generated as a polygraph by the atoms of its basis. In particular, maps out of it are determined by the images of these atoms [14, §3 before Definition 3.2, Theorem 6.1]. The concrete complexes needed here are as follows:

- (i) The normalized simplex complex $M[m]$ has degree- p basis $[i_0, \dots, i_p]$ for $0 \leq i_0 < \cdots < i_p \leq m$, with

$$d[i_0, \dots, i_p] = \sum_{j=0}^p (-1)^j [i_0, \dots, \widehat{i_j}, \dots, i_p] \quad (p > 0), \quad \epsilon[i] = 1.$$

It is strong Steiner and $\nu M[m] = \mathcal{O}_m$. A monotone map of ordinals acts by applying it to the tuple; a tuple with a repeated entry is zero. The atom of a nonempty tuple has its first and last vertices as its 0-source and 0-target.

- (ii) The globe complex $G[i]$ has one top basis element in degree i and two basis elements, its source and target faces, in each degree below i . Its positive-degree differentials are target minus source, and its vertices have augmentation one. It is strong Steiner and $\nu G[i] = D_i$. For $i = 0$ there is just one vertex.

The strong Steiner properties and tensor closure follow from [14, Examples 3.8–3.10]. The disk identification follows from [14, Example 4.7 and Theorem 5.11]. For the oriental identification and its cosimplicial action, including repeated entries, see [15, Theorem 3.2 and §4 before Theorem 4.1].

The tensor of augmented directed complexes uses the ordinary nonnegative chain-complex tensor, with separate augmentation:

$$d(x \otimes y) = dx \otimes y + (-1)^{|x|} x \otimes dy, \quad \epsilon(x \otimes y) = \epsilon(x)\epsilon(y) \quad (|x| = |y| = 0). \quad (2.3)$$

Its positive submonoid is generated by the tensors of positive elements. Tensor products of the indicated bases are the relevant positive bases.

We use the biclosed Gray tensor product on *all* strict ω -categories supplied by [6, Theorem A.15 and §A.16]. It has unit D_0 , preserves small colimits separately in each variable, and has canonical natural identifications

$$\nu K \otimes \nu L \cong \nu(K \otimes L) \quad (K, L \text{ strong Steiner}). \quad (2.4)$$

Naturality here is for every morphism between strong Steiner complexes, including maps sending basis elements to sums or zero. This is the completed extension theorem; the general extension is not being inferred from the incomplete proof of [14, Theorem 7.3]. In particular, preservation of colimits in this paper is a property of the Gray tensor, not a claimed preservation property of ν .

It follows that $D_i \otimes D_1$ is generated by the atoms indexed by pairs of globe basis elements, of dimension the sum of their dimensions. The same holds for $\mathcal{O}_m \otimes D_1$. Globe face inclusions and simplex face inclusions identify the corresponding based subcomplexes, and hence the corresponding atoms in these tensor products. One way to see the last assertion directly is to compute the positive and negative boundaries of a basis element within the based subcomplex: they are unchanged by its inclusion. The atom construction therefore gives the same cell in the larger complex.

Throughout, the interval is the *right* tensor factor. Write e_0, e_1 for the degree-zero basis elements of $M[1]$ and u for its degree-one element, so that $du = e_1 - e_0$. Also write $e_a : D_0 \rightarrow D_1$ for its endpoint functors and $\epsilon : D_1 \rightarrow D_0$ for the unique functor. The projection is

$$\pi_X = \text{id}_X \otimes \epsilon : X \otimes D_1 \longrightarrow X \otimes D_0 \cong X.$$

No interchange of the two tensor factors is used.

2.2 Right cylinders and truncation

Biclosedness supplies an ω -category $H(Y)$ and natural bijections

$$\text{Hom}_{\omega\text{Cat}}(X, H(Y)) \cong \text{Hom}_{\omega\text{Cat}}(X \otimes D_1, Y). \quad (2.5)$$

In the notation of [6, §A.18], $H(Y) = \text{Hom}_{\text{oplax}}(D_1, Y)$; the representing formula (2.5) fixes the convention. Evaluation at e_0, e_1 and precomposition with π_X give, respectively, the endpoints and constant cylinders. To extend cylinders through pushouts in $n\text{Cat}$, we will transpose them to functors into $H(Y)$. The next lemma shows that this construction stays within $n\text{Cat}$ when the target Y does.

Lemma 2.1 (Internal hom and reflection). *Let $n < \infty$ and let Y be a strict n -category. Then $H(Y)$ is a strict n -category. Consequently the reflection unit $X \rightarrow \tau_n X$ induces a natural bijection*

$$\mathrm{Hom}_{\omega\mathrm{Cat}}((\tau_n X) \otimes D_1, Y) \xrightarrow{\cong} \mathrm{Hom}_{\omega\mathrm{Cat}}(X \otimes D_1, Y). \quad (2.6)$$

This bijection respects endpoints and restriction along functors in the first variable. For $n = \infty$ the same assertion holds with $\tau_n = \mathrm{id}$.

Proof. We prove the special case needed here directly; the general internal-hom truncation assertion is also [6, Proposition A.29]. An i -cell of $H(Y)$ is a functor $P : D_i \otimes D_1 \rightarrow Y$. Let $s : D_{i-1} \rightarrow D_i$ be the source face inclusion, and let $p : D_i \rightarrow D_{i-1}$ send the top i -cell to the identity of the top $(i-1)$ -cell. Thus $ps = \mathrm{id}$ and $q = sp$ collapses D_i onto its source face. Under (2.5), the identity of the source of P is represented by $P(q \otimes \mathrm{id})$. We show that these functors agree when $i > n$.

The atoms of $D_i \otimes D_1$ outside the source face $D_{i-1} \otimes D_1$ come from exactly the following pairs: the top i -generator or the target $(i-1)$ -generator of D_i , paired with e_0, e_1 , or u . Their dimensions are respectively $i, i, i+1$ and $i-1, i-1, i$. If $i > n+1$, all have dimension greater than n . If $i = n+1$, only the target n -generator paired with e_0 or e_1 has dimension at most n . At either endpoint, the top $(n+1)$ -cell maps under P to an identity, so its source and target n -cells have the same image. These are precisely the two equalities needed to compare P with $P(q \otimes \mathrm{id})$ on the remaining degree- n atoms. All atoms in the source face are fixed by $q \otimes \mathrm{id}$.

The two functors therefore agree on all atoms of dimension at most n . They agree on higher atoms as well: in an n -category the image of a cell of dimension greater than n must be the iterated identity on its n -source, whose image has already been determined. Generation by atoms proves $P = P(q \otimes \mathrm{id})$. Every cell of $H(Y)$ above n is thus an identity.

Now transpose a cylinder to a functor $X \rightarrow H(Y)$. Since $H(Y)$ is an n -category, it factors uniquely through $\tau_n X$. Transposing back proves (2.6). Naturality of the reflection and of the adjunction proves the restriction assertion. The endpoint assertions follow either by naturality of evaluation or by uniqueness of the factorization into Y . $\square$

Remark 2.2. Even when X and Y are n -categories, the expression $X \otimes D_1$ denotes their tensor in $\omega\mathrm{Cat}$; it can have cells above n . A map from this tensor to Y is nevertheless represented by a map $X \rightarrow H(Y)$ in $n\mathrm{Cat}$. This is the fact that permits stationary cylinders to be glued by arbitrary pushouts in $n\mathrm{Cat}$.

2.3 The Alexander–Whitney map

For the Gray-cylinder-to-Street-nerve construction, see Ara [3, Sections 5.2.2–5.2.6, especially Proposition 5.2.3 and the right-cylinder formula in 5.2.6]. The verification needed here is included below; the classical Alexander–Whitney diagonal is credited separately in the proof.

Lemma 2.3 (A right Gray cylinder gives a nerve homotopy). *There is a natural simplicial map*

$$J_X : N_\infty X \times \Delta[1] \longrightarrow N_\infty(X \otimes D_1)$$

whose restriction at the endpoint a is $N_\infty(\mathrm{id}_X \otimes e_a)$, and for which $N_\infty(\pi_X)J_X$ is the projection to $N_\infty X$. Consequently a strict functor $P : X \otimes D_1 \rightarrow Y$ induces a simplicial homotopy between the nerves of its endpoint functors. If a functor $a : A \rightarrow X$ satisfies $P(a \otimes \mathrm{id}) = h\pi_A$ for some $h : A \rightarrow Y$, the homotopy restricts along $N_\infty a$ to the constant homotopy of $N_\infty h$. The same assertions apply to N_n when X, Y are n -categories.

Proof. For each m define the classical Alexander–Whitney diagonal [10, IV.2, Remark 2.5]

$$\text{AW}_m[i_0, \dots, i_p] = \sum_{a=0}^p [i_0, \dots, i_a] \otimes [i_a, \dots, i_p]. \quad (2.7)$$

It is a positive chain map $M[m] \rightarrow M[m] \otimes M[m]$. For clarity, in the differential of the right side, deleting a vertex strictly before or after the shared vertex gives the corresponding term in $\text{AW}_m d$, with coefficient $(-1)^j$ for deletion of i_j . The remaining terms cancel in pairs: deleting the final vertex of the first factor at cut a contributes

$$(-1)^a [i_0, \dots, i_{a-1}] \otimes [i_a, \dots, i_p],$$

whereas deleting the initial vertex of the second factor at cut $a - 1$ contributes the same tensor with coefficient $(-1)^{a-1}$. These cancellations use exactly the sign in (2.3). The degree-zero assertion is immediate. Every summand in (2.7) is positive. On vertices the augmentation is one; applying the augmentation to the first or second tensor factor selects, respectively, only the cut $a = 0$ or $a = p$. Thus both counits give the identity.

The diagonal is natural for every monotone ordinal map. For a tuple whose image is strictly increasing, this follows term by term. If the image has equal consecutive entries in positions $j, j + 1$, then at every cut $a \leq j$ the second factor is degenerate, and at every cut $a \geq j + 1$ the first factor is degenerate. Both sides of naturality are therefore zero after normalization. This verifies degeneracies as well as faces. Only this cosimplicial naturality is asserted.

Apply ν and (2.4) to obtain natural functors $\delta_m : \mathcal{O}_m \rightarrow \mathcal{O}_m \otimes \mathcal{O}_m$. For an m -simplex $F : \mathcal{O}_m \rightarrow X$ and a monotone map $\theta : [m] \rightarrow [1]$, put

$$J_X(F, \theta) = (F \otimes \mathcal{O}_\theta) \delta_m,$$

where $\mathcal{O}_\theta : \mathcal{O}_m \rightarrow \mathcal{O}_1 = D_1$ is the cosimplicial structure map. The naturality just checked makes J_X simplicial. If θ is constant at a , this map factors through the right counit and the endpoint e_a , giving the stated endpoint formula. Applying π_X also gives the right counit, so $N_\infty(\pi_X) J_X$ is the projection. Postcomposing with $N_\infty P$ proves the homotopy assertion, and naturality in X together with the projection formula proves stationarity. $\square$

2.4 Finiteness and gluing in simplicial sets

We use the Kan–Quillen model structure on $\mathbf{sSet}$. Its cofibrations are the monomorphisms, every object is cofibrant, and the model structure is proper. We also use the classical Kan replacement $K \rightarrow \text{Ex}^\infty K$, which is a weak equivalence and has Kan target. These are simplicial homotopy-theory inputs; they do not presuppose any model structure on strict higher categories [10, II.8.6 and III.4.8].

Lemma 2.4 (Presentability and filtered colimits). *For every $1 \leq n \leq \infty$, the category $n\mathbf{Cat}$ is locally finitely presentable, and filtered colimits are computed on cells. Each $\tau_n \mathcal{O}_m$ is finitely presentable in $n\mathbf{Cat}$. Consequently N_n , and also $\text{Ex}^r N_n$ for every finite $r \geq 0$, preserve filtered colimits. Simplicial weak equivalences are closed under filtered colimits of arrows, as are the maps f of $n\mathbf{Cat}$ for which $N_n f$ is a weak equivalence. No monomorphism hypothesis on transition maps is required.*

Proof. Strict n -categories are models of a small finitary essentially algebraic theory. The sorts are the cell dimensions; the operations are source, target, identity, and the binary compositions. A composition is defined by finitely many matching-boundary equations, and every strict category axiom is an equation involving finitely many cells. In the ω -case there are countably many sorts, but

all operations and equations still involve finite data. Proposition A.3 proves local finite presentability, completeness, cocompleteness and a common regular presentability bound directly from this signature. Its presentation construction includes composites made possible by new boundary identifications. The general essentially algebraic approach is developed in [1, 2]; for its finitary partial-operation formulation, see also [12, pp. 52–53].

For the cellwise colimit assertion one can argue directly. Form the filtered colimit of the sets of cells in each dimension. Any finite collection of representatives is represented at a common stage. If their boundaries agree in the colimit, those finitely many equalities hold at some later common stage, where the required composition is defined. Define its colimit value by that composition. A further common stage proves independence of choices, and each strict category law follows at a stage containing its finitely many entries. This constructs the colimit with its universal property. If all inputs are n -categories, every cell above n is already an identity, so the same is true of the colimit.

The oriental $\mathcal{O}_m$ has a finite polygraphic presentation: there is one generator for each nonempty strictly increasing tuple in $[m]$, and its boundary is an expression in lower generators. A map from $\mathcal{O}_m$ into a filtered colimit is thus specified by finitely many cell representatives satisfying finitely many boundary equations. These representatives and equations occur at one stage. Equality of two such maps is likewise detected at one stage. Hence $\mathcal{O}_m$ is finitely presentable in ωCat . The inclusion $n\text{Cat} \hookrightarrow \omega\text{Cat}$ preserves filtered colimits by the preceding cellwise calculation, so reflection and adjunction give finite presentability of $\tau_n\mathcal{O}_m$. This proves the assertion for N_n degreewise; compare also [4, Proposition 5.13(a)]. Since $(\text{Ex}K)_m = \text{Hom}_{\text{sSet}}(\text{Sd}\Delta[m], K)$ and $\text{Sd}\Delta[m]$ is a finite simplicial set, Ex preserves filtered colimits too.

Here are details of the weak-equivalence assertion. The functor $Q = \text{Ex}^\infty$ preserves filtered colimits, since Ex does and colimits commute with colimits. A filtered colimit of Kan complexes is Kan: any horn has finitely many nondegenerate simplices, so it and its compatibility equations occur at one stage and can be filled there. For a filtered diagram of Kan complexes K_i , components commute with its colimit, since vertices and finite paths are finite data. If x is a vertex of $\text{colim}_i K_i$, choose a representative $x_i \in K_i$. The coslice of arrows $i \rightarrow j$ is filtered and its projection to the indexing category is final; it gives a diagram with compatible basepoints x_j . For every $k \geq 1$,

$$\text{colim}_{(i \rightarrow j)} \pi_k(K_j, x_j) \cong \pi_k(\text{colim}_j K_j, x).$$

Indeed, use the finite pointed sphere $\Delta[k]/\partial\Delta[k]$ and its finite reduced simplicial cylinder to represent classes and pointed homotopies in a Kan complex. Their simplices and all required equalities, including the basepoint equality, occur at a sufficiently late stage. This proves both surjectivity and injectivity of the displayed map, including eventual equalities when transition maps are not injective. It follows that an objectwise weak equivalence between filtered diagrams of Kan complexes induces a weak equivalence of colimits. Apply this to Q of a diagram of arbitrary simplicial weak equivalences, and use the natural weak equivalences $K \rightarrow QK$ and two-out-of-three. Finally apply preservation of filtered colimits by N_n . $\square$

Lemma 2.5 (Gluing with one monomorphic leg). *Consider a commutative map of spans of simplicial sets*

$$\begin{array}{ccccc} A & \longleftarrow & C & \xleftarrow{i} & B \\ a \downarrow & & c \downarrow & & \downarrow b \\ A' & \longleftarrow & C' & \xleftarrow{i'} & B' \end{array}$$

where i and i' are monomorphisms. If a, b, c are weak equivalences, then the canonical map

$$A \amalg_C B \longrightarrow A' \amalg_{C'} B'$$

is a weak equivalence. There is no condition on the maps $C \rightarrow A$ and $C' \rightarrow A'$. Each of the displayed ordinary pushouts computes a homotopy pushout. In particular, cobase change of a weak equivalence along a simplicial monomorphism is a weak equivalence.

Proof. This is the simplicial gluing lemma [10, II.8.8 and II.8.12]; the following proof keeps track of the specified cofibrations. Put

$$E = B \amalg_C C'.$$

By simplicial left properness, $B \rightarrow E$ is a weak equivalence, since it is the pushout of c along the monomorphism i . The induced map $E \rightarrow B'$ is consequently a weak equivalence by two-out-of-three. Both $C' \rightarrow E$ and $C' \rightarrow B'$ are cofibrations. They are therefore cofibrant objects of the undercategory $C'/\mathbf{sSet}$, with its inherited model structure. Cobase change along $C' \rightarrow A'$ is a left Quillen functor

$$C'/\mathbf{sSet} \longrightarrow A'/\mathbf{sSet} : (C' \rightarrow T) \longmapsto (A' \rightarrow A' \amalg_{C'} T).$$

Indeed, its right adjoint is restriction of the structure map, which preserves fibrations and trivial fibrations. It preserves weak equivalences between cofibrant objects, so

$$A' \amalg_C B \cong A' \amalg_{C'} E \longrightarrow A' \amalg_{C'} B'$$

is a weak equivalence. Also $A \rightarrow A \amalg_C B$ is a monomorphism, being a pushout of i . Left properness applied to a gives a weak equivalence $A \amalg_C B \rightarrow A' \amalg_C B$. Composing the two proves the result.

To see the homotopy-pushout assertion, factor the possibly nonmonomorphic map $C \rightarrow A$ as a cofibration $C \hookrightarrow \tilde{A}$ followed by a trivial fibration $\tilde{A} \rightarrow A$. The span $\tilde{A} \leftarrow C \hookrightarrow B$ is a cofibrant replacement of the original span for calculating the derived pushout: its three objects are cofibrant and both legs are cofibrations. The already proved assertion identifies its pushout with $A \amalg_C B$ up to weak equivalence. For the final assertion, one may either use left properness directly or apply the span assertion to $X \leftarrow X \hookrightarrow Z$ and $Y \leftarrow X \hookrightarrow Z$, with vertical maps $X \rightarrow Y, \text{id}_X, \text{id}_Z$. $\square$

3 Positive relative prisms

Throughout this section the interval is the *right* tensor factor. At the chain level this is the antihomotopy convention of [6, Appendix B.4.1–B.4.2]. Gagna constructs simplicial homotopies by positive concatenation of chains under an order condition on the endpoint maps [9, Sections 6.3–6.8]. We use a related positive formula under the sieve and constancy hypotheses below to construct a strict right Gray cylinder stationary on the entire old subcategory. When its endpoint maps fix that subcategory, it glues to the constant cylinder on an arbitrary attaching target. Write $\epsilon : D_1 \rightarrow D_0$ for its unique map to the Gray unit, and write

$$\pi_B = \text{id}_B \otimes \epsilon : B \otimes D_1 \longrightarrow B.$$

For $i : A \rightarrow B$ and $v : A \rightarrow C$, a cylinder $P : B \otimes D_1 \rightarrow C$ is *stationary along i relative to v* if $P(i \otimes \text{id}) = v\pi_A$. This is an equality of strict functors, including every cell of A . The special prism below supplies this equality even when the common endpoint map on A is not the inclusion.

Lemma 3.1 (Relative prism). *Let $A \subset E$ be a downward-closed subposet, and let $f, g : E \rightarrow E$ be monotone maps such that*

$$f(e) \leq g(e) \quad (e \in E), \quad f|_A = g|_A, \quad g|_{E \setminus A} \text{ is constant.}$$

The constancy condition is vacuous when $E \setminus A$ is empty. For each $1 \leq n \leq \infty$ there is a strict functor of ω -categories

$$P_{f,g} : U_E \otimes D_1 \longrightarrow U_E$$

whose restrictions to the two interval endpoints are $U(f)$ and $U(g)$. If $i : A \hookrightarrow E$ and $v = fi = gi : A \rightarrow E$, then

$$P_{f,g}(U(i) \otimes \text{id}_{D_1}) = U(v)\pi_{U_A}. \quad (3.1)$$

Consequently its endpoints induce simplicially homotopic maps on Street nerves, with the homotopy stationary relative to $N_n U(v)$.

Proof. Simplexwise chain map. We first construct the cylinder in ωCat , before applying τ_n . If T is a nonempty finite totally ordered set, let $M[T]$ be its augmented normalized simplex chain complex. Thus its positive basis in degree p consists of the strictly increasing tuples $[t_0, \dots, t_p]$. A weakly increasing tuple with a repeated entry means zero. For $p > 0$ the differential is the alternating sum of deletions; the augmentation sends every vertex to 1. The alternating deletion formula remains valid for a tuple with repetitions: it is the differential in the normalized quotient of the unnormalized simplex complex. In particular all calculations below remain valid when a displayed tuple is zero.

Fix a simplex $\sigma = (v_0 \leq \dots \leq v_m)$ of NE and put

$$T_\sigma = \{f(v_i), g(v_i) : 0 \leq i \leq m\} \subset E.$$

This subset is totally ordered. Indeed, the indices whose vertices belong to A form a prefix. On that prefix the two image lists coincide. If there are remaining vertices, all their g -images are one element b , and every $f(v_i)$ is at most $f(v_m) \leq b$. Consequently T_σ consists of the ordered f -image list, possibly with this one upper element adjoined. If every vertex is old, T_σ is just the f -image list. The order embedding $T_\sigma \hookrightarrow E$ gives the canonical map

$$\nu M[T_\sigma] \longrightarrow U_E^\infty.$$

Here $\nu M[T_\sigma]$ is the oriental obtained by enumerating T_σ increasingly.

Let e_0, e_1 denote the vertices and u the edge of $M[1]$, so $du = e_1 - e_0$. We define a graded homomorphism

$$C_\sigma : M[m] \otimes M[1] \longrightarrow M[T_\sigma] \quad (3.2)$$

on the tensor basis. For $x = [i_0, \dots, i_p]$, with $i_0 < \dots < i_p$, set $w_j = v_{i_j}$ and prescribe

$$\begin{aligned} C_\sigma(x \otimes e_0) &= F(w) := [f(w_0), \dots, f(w_p)], \\ C_\sigma(x \otimes e_1) &= G(w) := [g(w_0), \dots, g(w_p)], \\ C_\sigma(x \otimes u) &= h(w) := [f(w_0), \dots, f(w_p), g(w_p)]. \end{aligned} \quad (3.3)$$

Each image is a positive basis element or zero: all displayed tuples are weakly increasing in T_σ . Both endpoint prescriptions are the normalized chain maps of monotone maps on vertices. They preserve augmentation, and hence so does C_σ in degree zero.

It remains to check the differential on $x \otimes u$, including its sign. If $p = 0$, the required identity is precisely

$$d[f(w_0), g(w_0)] = [g(w_0)] - [f(w_0)].$$

For $p > 0$, abbreviate

$$\begin{aligned} B &= [f(w_0), \dots, f(w_{p-1}), g(w_p)], \\ B' &= [f(w_0), \dots, f(w_{p-1}), g(w_{p-1})]. \end{aligned}$$

We claim that $B - B' = G(w)$ in normalized chains. If $w_{p-1} \in A$, all $w_0, \dots, w_{p-1}$ lie in A , so that $B = G(w)$, whereas $B' = 0$ because its last two entries coincide. If $w_{p-1} \notin A$, then also $w_p \notin A$. Constancy of g on the complement gives $B = B'$ and makes the last two entries of $G(w)$ coincide, so again $B - B' = G(w)$. Expanding $dh(w)$ and canceling the terms obtained by deleting one of $f(w_0), \dots, f(w_{p-1})$ therefore gives

$$\begin{aligned} dh(w) &= \sum_{j=0}^p (-1)^j h(w_0, \dots, \widehat{w_j}, \dots, w_p) + (-1)^p (B - B' - F(w)) \\ &= \sum_{j=0}^p (-1)^j h(w_0, \dots, \widehat{w_j}, \dots, w_p) + (-1)^p (G(w) - F(w)). \end{aligned} \quad (3.4)$$

This is exactly the image under C_σ of

$$d(x \otimes u) = dx \otimes u + (-1)^p x \otimes (e_1 - e_0).$$

Thus (3.2) is a positive, augmentation-preserving chain map in every degree. The strong Steiner tensor comparison gives the strict functor

$$\mathcal{O}_m \otimes D_1 \cong \nu(M[m] \otimes M[1]) \xrightarrow{\nu C_\sigma} \nu M[T_\sigma] \longrightarrow U_E^\infty. \quad (3.5)$$

Compatibility with simplex operators. We next verify compatibility with *all* simplex operators. Let $\alpha : [q] \rightarrow [m]$ be monotone and set $\rho = \sigma\alpha$. There is an order embedding $j : T_\rho \hookrightarrow T_\sigma$. We have an equality of whole directed chain maps

$$M(j)C_\rho = C_\sigma(M(\alpha) \otimes \text{id}_{M[1]}). \quad (3.6)$$

To check it, take a basis tuple $[a_0, \dots, a_p]$ in $M[q]$. If α is injective on this tuple, both sides of (3.6), on tensoring with e_0 , e_1 , or u , are exactly the same tuple in (3.3), selected by the indices $\alpha(a_0), \dots, \alpha(a_p)$. If α is not injective on it, the right side is zero by normalization. On the left, equal selected indices yield repeated f -images in the e_0 and u prescriptions, and repeated g -images in the e_1 prescription, so all three images are zero. These cases include every face, every degeneracy, and every composite of them, also when σ itself has repeated vertices. Naturality of the strong Steiner comparison and functoriality of ν turn (3.6) into an equality of the strict functors (3.5) after mapping into U_E^∞ .

The zero chains in this argument retain their lower boundary data after applying ν . In the cell table (2.2), the entries satisfy $dx_r^\pm = x_{r-1}^+ - x_{r-1}^-$. If the top entry of a q -cell is zero, with $q > 0$, its two $(q-1)$ -entries therefore coincide; the resulting cell is the identity on that common $(q-1)$ -cell. That lower cell need not be an identity. For example, take $E = \{a < b\}$, $A = \{a\}$, f constant at a , and $g = \text{id}$. The degree-two tensor generator has image $h(a, b) = [a, a, b] = 0$, but its associated square maps to the identity 2-cell on the nonidentity edge $[a, b]$. Equation (3.6) compares every degree of the chain maps, so it establishes agreement on these retained boundaries as well.

Assembly and stationarity. The oriental presentation and cocontinuity of the Gray tensor in its first variable now give

$$U_E^\infty \otimes D_1 \cong \text{colim}_{\sigma: \Delta[m] \rightarrow NE} (\mathcal{O}_m \otimes D_1).$$

The compatible maps (3.5) consequently induce $P_{f,g}^\infty : U_E^\infty \otimes D_1 \rightarrow U_E^\infty$. This colimit is taken in strict ω -categories; the argument uses tensor cocontinuity, not preservation of arbitrary colimits by ν . The two endpoint maps are $U^\infty(f)$ and $U^\infty(g)$ by their simplexwise formulas.

On a simplex entirely in A , the maps F and G coincide and $h(w) = 0$ for every selected tuple, because $f(w_p) = g(w_p)$. More precisely, the entire chain map is

$$C_\sigma = F \circ (\text{id}_{M[m]} \otimes \epsilon), \quad \epsilon(e_0) = \epsilon(e_1) = 1, \quad \epsilon(u) = 0,$$

where the chain augmentation of the interval is denoted by the same letter as its categorical realization. Applying ν and the simplex colimit proves strict stationarity on U_A^∞ relative to $U^\infty(v)$. It is this factorization, rather than the vanishing of individual top chains alone, that proves stationarity.

Finite-dimensional reflection. For finite n , let $q_E : U_E^\infty \rightarrow U_E$ be the reflection unit. By Lemma 2.1, the composite $q_E P_{f,g}^\infty$ factors uniquely as

$$U_E^\infty \otimes D_1 \xrightarrow{q_E \otimes \text{id}} U_E \otimes D_1 \xrightarrow{P_{f,g}} U_E.$$

Its endpoint identities follow by precomposing with q_E and using the universal property of reflection. To check (3.1), precompose both sides with $q_A \otimes \text{id}$. The resulting equality is the stationarity just proved before reflection. The injectivity in the natural bijection of Lemma 2.1, with target U_E , gives the claimed equality after reflection. For $n = \infty$ no reflection is needed. If E or A is empty, the corresponding colimit is empty and the same statements hold with the unique maps from it. Finally Lemma 2.3 gives the stated simplicial homotopy and its stationarity. $\square$

Lemma 3.2 (Gluing a stationary cylinder). *Let $i : B_0 \rightarrow B$ and $a : B_0 \rightarrow X$ be arbitrary strict functors of n -categories, and form the pushout*

$$\begin{array}{ccc} B_0 & \xrightarrow{i} & B \\ a \downarrow & & \downarrow b \\ X & \xrightarrow{j} & Y. \end{array}$$

Suppose $F_0, F_1 : B \rightarrow B$ satisfy $F_0 i = F_1 i = i$, and suppose $P : B \otimes D_1 \rightarrow B$ has endpoints F_0, F_1 and satisfies $P(i \otimes \text{id}) = i \pi_{B_0}$. Then there is a cylinder $Q : Y \otimes D_1 \rightarrow Y$ stationary on X , whose endpoints $\bar{F}_0, \bar{F}_1 : Y \rightarrow Y$ are specified by

$$\bar{F}_\delta j = j, \quad \bar{F}_\delta b = b F_\delta \quad (\delta = 0, 1).$$

Its restriction along $b \otimes \text{id}$ is bP . This construction is natural in $a : B_0 \rightarrow X$, and its nerve gives a simplicial homotopy between $N_n \bar{F}_0$ and $N_n \bar{F}_1$ stationary along $N_n j$. In particular it applies to Lemma 3.1 whenever $f|_A = g|_A = \text{id}_A$, for every attaching functor $U_A \rightarrow X$.

Proof. Let $H(Y)$ be the right internal hom representing strict functors $(-) \otimes D_1 \rightarrow Y$. It is an n -category by Lemma 2.1; for $n = \infty$ this is automatic. The cylinders

$$bP : B \otimes D_1 \rightarrow Y, \quad j\pi_X : X \otimes D_1 \rightarrow Y$$

transpose to maps $B \rightarrow H(Y)$ and $X \rightarrow H(Y)$ in $n\text{Cat}$. Their restrictions to B_0 agree, since on its cylinder both give

$$bi\pi_{B_0} = ja\pi_{B_0}.$$

The pushout universal property in $n\text{Cat}$ thus gives a unique map $Y \rightarrow H(Y)$ extending both transposes. Its transpose is Q . The endpoint formulas, the equality $Q(j \otimes \text{id}) = j\pi_X$, and the restriction to B follow from naturality of the adjunction and the same pushout universal property. For a map $X \rightarrow X'$ commuting with the attaching maps, the induced map $Y \rightarrow Y'$ carries this cylinder to the corresponding cylinder: the two maps $Y \rightarrow H(Y')$ agree on X and B and hence agree. This proves naturality. Lemma 2.3 proves the final homotopy assertion. No injectivity, freeness, or cofibrancy hypothesis on i , a , or X entered this construction. $\square$

4 Labelled pieces of algebraic pushouts

We fix $1 \leq n \leq \infty$ throughout this section. The arguments concern actual subcategories and actual pushouts. This precision will allow us to apply the simplicial gluing lemma even when an attaching functor identifies objects or cells.

The sieve and nerve constructions have established antecedents in [4, Propositions 5.3 and 5.5]. We give the calculations explicitly to identify the labels, fibers and intersections after an arbitrary attachment.

A full sub- n -category of C on a set of objects S has exactly those objects and contains every positive-dimensional cell whose 0-source and 0-target belong to S . It is a *sieve* if a 1-cell $x \rightarrow y$ with $y \in S$ forces $x \in S$, and a *cosieve* if a 1-cell $x \rightarrow y$ with $x \in S$ forces $y \in S$. A sieve is equivalently the fiber over 0 of a strict functor $C \rightarrow D_1$; a cosieve is equivalently the fiber over 1. Indeed, the indicated object assignment extends uniquely to a strict functor: each nonempty hom in the target is terminal, and the sieve or cosieve condition excludes an arrow from a 1-labelled object to a 0-labelled object. These fibers are full, since all cells between objects with the same label necessarily map to identities at that label.

The following fiber adjoints are those of [4, Lemma 2.8].

Lemma 4.1 (Fibers and objects preserve colimits). *For $\epsilon \in \{0, 1\}$, the functor*

$$F_\epsilon : n\text{Cat}/D_1 \longrightarrow n\text{Cat}, \quad (C \xrightarrow{p} D_1) \longmapsto C_\epsilon = p^{-1}(\epsilon),$$

preserves all small colimits. The object functor $\text{Ob} : n\text{Cat} \rightarrow \text{Set}$ also preserves all small colimits. Both assertions hold directly for $n = \infty$.

Proof. We construct a right adjoint Q_0 to F_0 . Given B , put all its objects over 0 and adjoin a new object $*$ over 1. Keep the hom categories $B(b, b')$, and set

$$Q_0(B)(b, *) = \mathbf{1}, \quad Q_0(B)(*, b) = \emptyset, \quad Q_0(B)(*, *) = \mathbf{1}.$$

Here the homs are strict $(n-1)$ -categories for finite n , sets when $n = 1$, and strict ω -categories when $n = \infty$; $\mathbf{1}$ and $\emptyset$ denote their terminal and initial objects. Compositions inside B are retained. Every remaining composition either has empty domain or has terminal codomain, so is uniquely determined. Associativity, units, and interchange hold by the laws in B or by uniqueness. This defines $Q_0(B)$ and its map to D_1 in all the stated dimensions.

For $p : C \rightarrow D_1$, a functor $C_0 \rightarrow B$ extends uniquely over D_1 to $C \rightarrow Q_0(B)$. All objects of C_1 must go to $*$; their internal cells and the cells from C_0 to C_1 have unique images in the specified terminal homs. There are no cells from C_1 to C_0 , since their 1-boundaries would map to an arrow $1 \rightarrow 0$. Restriction and extension are inverse and natural, giving

$$\text{Hom}_{n\text{Cat}/D_1}(C, Q_0(B)) \cong \text{Hom}_{n\text{Cat}}(C_0, B).$$

Thus F_0 is a left adjoint. For F_1 , place B over 1 and adjoin a new initial object over 0, with terminal homs from that object into B , empty reverse homs, and terminal endomorphism hom. The same restriction argument gives a right adjoint Q_1 . In particular these constructions include $B = \emptyset$ and prove preservation of the empty colimit as well.

For a set S , let $K(S)$ be the ordinary chaotic category on S , with one arrow for each ordered pair of objects, regarded as an n -category with identity cells above dimension one. Every object function $\text{Ob} C \rightarrow S$ has a unique strict extension $C \rightarrow K(S)$. Consequently Ob is left adjoint to K and preserves colimits. $\square$

The next elementary property of orientals is needed before taking a colimit of simplex pieces. It is not enough merely to know their sets of vertices.

Lemma 4.2 (Full oriental segments). *Let I be an initial or a final segment of $[m] = \{0, \dots, m\}$. If I is nonempty and has $q + 1$ vertices, the face inclusion identifies $\tau_n \mathcal{O}_q$ with the full sub- n -category of $\tau_n \mathcal{O}_m$ on I . If I is empty, that full subcategory is empty. Here and below $\tau_n = \text{id}$ when $n = \infty$.*

Proof. Recall the oriental presentation from Section 2: its atoms are indexed by strictly increasing tuples, with 0-source the first vertex and 0-target the last vertex. All cells are finite composites and identities of these atoms. In particular every 1-cell has nondecreasing endpoints.

Suppose a cell has 0-source a and 0-target b . In any expression for it, every occurring atom has all its vertices in $[a, b]$. This follows by induction on the expression. For an atom it is the tuple description. In a 0-composite the factors have endpoints (a, c) and (c, b) , with $a \leq c \leq b$. In a composition in positive dimension both factors have the same 0-endpoints as the composite. Identities preserve these endpoints, and an identity on a vertex has support at that vertex. These observations prove the induction. If $a, b \in I$, the whole interval $[a, b]$ lies in I , so the expression uses only the atoms of that face. Thus the face generates every cell on those objects.

For finite $n \geq 1$, reflection leaves objects unchanged and every cell of dimension at most n is represented by a cell before reflection. The same argument applied to a representative proves surjectivity onto the full subcategory after reflection; higher cells are identities. For injectivity, the inclusion $I \hookrightarrow [m]$ has a monotone retraction: clamp to the last vertex of I for an initial segment, or to its first vertex for a final segment. By cosimplicial functoriality it induces a retraction on orientals, and then on their reflections. The face map is therefore injective on every cell set. This proves the full-subcategory assertion. If I is empty there are no objects and hence no cells; no retraction is needed. $\square$

Lemma 4.3 (Poset pieces). *There is a canonical identification $\text{Ob } U_P = P$ for every poset P . If $S \subset P$ is a sieve, the map $U_S \rightarrow U_P$ is the inclusion of the full 0-fiber of the membership label $U_P \rightarrow D_1$. If S is a cosieve, it is the inclusion of the full 1-fiber. These identifications are natural for maps of labelled posets and include empty posets and empty fibers.*

Proof. Use the canonical simplex presentation

$$U_P = \text{colim}_{\sigma: \Delta[m] \rightarrow NP} \tau_n \mathcal{O}_m. \quad (4.1)$$

By Lemma 4.1, its object set is the colimit of the vertex sets $[m]$. Sending the j th vertex of a simplex σ to $\sigma(j)$ identifies this colimit with P : each element occurs as a zero-simplex, and the face inclusions from zero-simplices identify every vertex with that occurrence.

For a sieve S , its characteristic map $P \rightarrow [1]$ assigns 0 on S and 1 on its complement. It is monotone and induces the displayed label on U_P , since $U_{[1]} = D_1$. The presentation (4.1) is consequently a colimit in $n\text{Cat}/D_1$ as well: slice colimits have the underlying colimits in $n\text{Cat}$. In its σ -piece the vertices mapping to S form an initial segment I_σ . By Lemma 4.2, the 0-fiber of that piece is its truncated segment oriental. Taking fibers commutes with the colimit by Lemma 4.1.

We verify that this colimit of segment orientals is canonically U_S . Restricting each nonempty simplex segment to S gives a map from its segment oriental to U_S ; the empty segment has the unique such map. These maps are compatible with every simplex operator. In the reverse direction, each simplex entirely in S supplies its whole oriental to the fiber colimit. The two composites are identities. On a simplex of S this is immediate. On a nonempty segment of an arbitrary simplex σ , it follows from the face inclusion selecting I_σ , which is a morphism in the simplex category of NP ;

on an empty segment there is nothing to check. The isomorphism so obtained is the canonical map $U_S \rightarrow U_P$, followed into its fiber. Fibers are full sieves, which proves the claim. The argument for a cosieve uses label 1 on S , final vertex segments, and F_1 throughout. All constructions commute with maps preserving the labels. $\square$

Here is the form of the fiber calculation that we will use in pushouts. It also records precisely which hypotheses are needed of a label on an arbitrary attaching target.

Proposition 4.4 (Labelled attachments). *Let $A \subset P$ be a full subposet, let $a : U_A \rightarrow X$ be any strict functor, and form*

$$Y = X \amalg_{U_A} U_P.$$

Suppose a monotone map $\lambda : P \rightarrow [1]$ and a strict functor $\xi : X \rightarrow D_1$ satisfy

$$\xi a = U(\lambda|_A) : U_A \longrightarrow D_1, \quad U_{[1]} = D_1.$$

Write $P_\epsilon = \lambda^{-1}(\epsilon)$, $A_\epsilon = A \cap P_\epsilon$, and $X_\epsilon = \xi^{-1}(\epsilon)$. Then the induced label $Y \rightarrow D_1$ has actual full fibers

$$Y_\epsilon \cong X_\epsilon \amalg_{U_{A_\epsilon}} U_{P_\epsilon}. \quad (4.2)$$

Moreover, independently of these labels,

$$\text{Ob } Y \cong \text{Ob } X \amalg_A P \cong \text{Ob } X \amalg (P \setminus A). \quad (4.3)$$

For two such pairs of compatible labels, write $\bar{\lambda}_1, \bar{\lambda}_2 : Y \rightarrow D_1$ for the induced labels. Their 0-fibers intersect in the 0-fiber of $\max \circ (\bar{\lambda}_1, \bar{\lambda}_2)$; their 1-fibers intersect in the 1-fiber of $\min \circ (\bar{\lambda}_1, \bar{\lambda}_2)$. Formula (4.2), using the corresponding maximum or minimum on both original legs, computes these intersections as well.

Proof. The compatibility assumption makes the defining pushout a pushout in $n\text{Cat}/D_1$. Applying F_ϵ and then Lemmas 4.1 and 4.3 gives (4.2). This identifies the canonical map from the displayed pushout to Y with the inclusion of a full fiber. The object formula follows from the object part of Lemma 4.1 and Lemma 4.3. Its last isomorphism uses only the injectivity of $A \hookrightarrow P$: old vertices of P are replaced by their images in X , while vertices in $P \setminus A$ remain distinct new objects. There is no injectivity requirement on $\text{Ob } a$.

Two labels on Y give a functor $Y \rightarrow D_1 \times D_1$. The monotone functions $\max, \min : [1] \times [1] \rightarrow [1]$ are strict functors, so their composites are labels on Y . An object has maximum label 0 exactly when both labels are 0, and minimum label 1 exactly when both labels are 1. Full subcategories on the same objects are equal, which proves the assertions about intersections. The composite labels restrict to the corresponding maximum or minimum labels on both legs of the original pushout, so the same fiber formula applies. $\square$

For use in the next two sections, we spell out both cover consequences.

Corollary 4.5 (Sieve and cosieve covers after attachment). *With the notation of Proposition 4.4, the following are canonical identifications of subcategories of Y .*

(i) *Suppose P_1, P_2 are sieves of P , contain A , and cover P . Then the subcategories*

$$Y_i = X \amalg_{U_A} U_{P_i} \quad (i = 1, 2)$$

are sieves covering Y on objects, and

$$Y_1 \cap Y_2 = X \amalg_{U_A} U_{P_1 \cap P_2}.$$

(ii) Suppose T_1, T_2 are cosieves of P , cover P , satisfy $A \subset T_1$, and satisfy $A \cap T_2 = \emptyset$. Then

$$Y_1 = X \amalg_{U_A} U_{T_1}, \quad Y_2 = U_{T_2}$$

are cosieves covering Y on objects, and $Y_1 \cap Y_2 = U_{T_1 \cap T_2}$.

If A itself is a sieve of P , the canonical map $X \rightarrow Y$ is the inclusion of a full sieve.

Proof. For (i), label membership in each P_i by 0 and label every object of X by 0. Apply Proposition 4.4 to the two labels and their maximum. For (ii), label membership in each T_i by 1; use constant label 1 on X for T_1 and constant label 0 on X for T_2 . These agree on the attaching source by the stated conditions. The 1-fiber of the second old-side label is empty, and so is that of the minimum of the two old-side labels. The same is true on their attaching sources. Formula (4.2) therefore gives U_{T_2} and $U_{T_1 \cap T_2}$ exactly. In both cases (4.3) proves the object cover. Finally, when A is a sieve, label A and X by 0 and $P \setminus A$ by 1. The 0-fiber formula is $X \amalg_{U_A} U_A = X$. $\square$

In particular, an attaching functor may identify two old objects, collapse an old arrow to an identity, or identify parallel old cells. These operations cause no ambiguity in the cover calculations: the labels agree on every old cell before the pushout is taken. The fiber adjunctions compute the resulting subcategories after all those identifications.

Lemma 4.6 (Street nerves of two-piece covers). *Let $C_1, C_2 \subset C$ be two full sieves which cover the objects of C , or two full cosieves which cover its objects. Put $C_{12} = C_1 \cap C_2$. The square*

$$\begin{array}{ccc} N_n C_{12} & \hookrightarrow & N_n C_2 \\ \downarrow & & \downarrow \\ N_n C_1 & \hookrightarrow & N_n C \end{array}$$

is an ordinary pushout of simplicial sets. All four maps are monomorphisms, so it is also a homotopy pushout in the Kan–Quillen model structure.

Proof. A full inclusion induces a monomorphism of Street nerves, and the intersection of the two nerves is $N_n C_{12}$: a functor whose image belongs to both subcategories has a unique factorization through their full intersection. It remains to prove that every simplex $z : \mathcal{O}_m \rightarrow C$ belongs to one of the two nerves. In the sieve case choose a piece containing $z(m)$. For each vertex $j \leq m$, the oriental has an arrow $j \rightarrow m$, whose image shows that $z(j)$ belongs to that piece. Fullness then puts every cell of the simplex in the same piece. In the cosieve case choose a piece containing $z(0)$ and use the arrows $0 \rightarrow j$. This includes $m = 0$ and proves the union statement degree by degree. A union of simplicial subsets is their ordinary pushout over the intersection. Since its legs are monomorphisms, the homotopy-pushout assertion follows from Lemma 2.5. $\square$

5 Couniversality and the unit on posets

Fix $n \in \{1, 2, \dots\} \cup \{\infty\}$ throughout this section. A *Thomason equivalence* is a strict functor whose Street nerve is a weak homotopy equivalence. This terminology uses no model structure on $n\text{Cat}$. We prove that certain poset inclusions remain Thomason equivalences after every pushout, and then establish the adjunction unit on posets independently of transfer.

For an inclusion $i : A \hookrightarrow E$ of posets, a right adjoint retraction $r : E \rightarrow A$ satisfies

$$a \leq e \iff a \leq r(e) \quad (a \in A, e \in E), \quad ri = \text{id}_A. \quad (5.1)$$

We identify A with its image in E . In particular $r(e) \leq e$. The inclusion is a *sieve* when $e \leq a \in A$ implies $e \in A$. For an arbitrary strict functor $a : U_A \rightarrow X$, write

$$Y = X \amalg_{U_A} U_E, \quad j : X \longrightarrow Y, \quad v : U_E \longrightarrow Y. \quad (5.2)$$

The symbol a here denotes the attaching functor; elements of A will be denoted by b below.

Lemma 5.1 (A greatest new element). *Suppose that $A \hookrightarrow E$ is a sieve with a right adjoint retraction, and that $E \setminus A$ has a greatest element. For every attaching functor in (5.2), the map j admits a retraction $p : Y \rightarrow X$ such that $N_n(jp)$ and $\text{id}_{N_n Y}$ are joined by a zigzag of simplicial homotopies. Consequently j is a Thomason equivalence. Neither A nor E needs to be finite.*

Proof. Let x be greatest in $E \setminus A$ and put $t = r(x) \in A$. If $b \in A$, $y \in E \setminus A$, and $b \leq y$, then $b \leq y \leq x$, so (5.1) gives $b \leq t$. Define maps $k, g : E \rightarrow E$ by

$$k(e) = \begin{cases} e, & e \in A, \\ t, & e \notin A, \end{cases} \quad g(e) = \begin{cases} e, & e \in A, \\ x, & e \notin A. \end{cases}$$

Both are monotone. Indeed, for a comparable pair $e \leq e'$, the old–old case is unchanged and the new–new case is constant. In the old–new case the preceding observation gives $e \leq t \leq x$. The new–old case cannot occur because A is a sieve. We therefore have pointwise inequalities

$$\text{id}_E \leq g, \quad k \leq g, \quad (5.3)$$

and all three maps restrict to id_A . The common upper map g is constant on $E \setminus A$, exactly as required in Lemma 3.1. Figure 1 shows why this common upper map is useful: id_E and k need not be pointwise comparable, although both lie below g .

Apply Lemma 3.2 to the two comparisons in (5.3). Let $G, K : Y \rightarrow Y$ be the pushout endomorphisms whose restrictions to X are j and whose restrictions to U_E are $v c_n N(g)$ and $v c_n N(k)$, respectively. The resulting right Gray cylinders, both stationary on X , have endpoint pattern

$$\text{id}_Y \longrightarrow G \longleftarrow K.$$

By Lemma 2.3, these give simplicial homotopies with the same endpoint pattern on Street nerves.

The image of k lies in A , so $k = i\bar{k}$ for a monotone retraction $\bar{k} : E \rightarrow A$. The maps id_X and $a c_n N(\bar{k}) : U_E \rightarrow X$ agree on U_A and hence define $p : Y \rightarrow X$. They give $jp = \text{id}_X$. Comparing restrictions to the two pushout summands gives $jp = K$: on X both restrict to j , and on U_E one has

$$jp v = j a c_n N(\bar{k}) = v c_n N(i) c_n N(\bar{k}) = v c_n N(k).$$

After realization, concatenate the homotopy from id_Y to G with the reverse of the homotopy from jp to G . Thus $|N_n p|$ is a homotopy inverse to $|N_n j|$, proving the assertion. This reversal is a reversal of an ordinary topological homotopy; no reversed Gray cylinder is asserted. $\square$

Theorem 5.2 (Couniversality). *If $i : A \hookrightarrow E$ is a sieve of small posets with a right adjoint retraction, then $c_n N(i)$ is a couniversal Thomason equivalence: for every $a : U_A \rightarrow X$, the canonical map*

$$X \longrightarrow X \amalg_{U_A} U_E$$

is a Thomason equivalence. This holds for every finite $n \geq 1$ and for $n = \infty$.

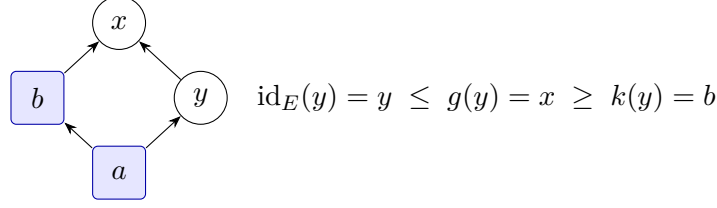


Figure 1: The shaded square nodes form the old sieve $A = \{a, b\}$. Its right adjoint retraction has $r(y) = a$ and $r(x) = t = b$. Thus $k(y) = b$ is incomparable with y ; the common upper value x supplies the two comparisons used by the relative prisms.

Proof. First suppose that E is finite. We use induction on $|E \setminus A|$, simultaneously for all attaching targets X and all attaching functors a . When this cardinality is zero, j is an isomorphism. When the complement has a greatest element, Lemma 5.1 applies.

Otherwise the nonempty finite poset $E \setminus A$ has at least two distinct maximal elements x_1, x_2 : a finite poset with a unique maximal element has that element as its greatest element. Each x_i is also maximal in E , since an element above a new element cannot belong to the sieve A . Put

$$E_1 = E \setminus \{x_1\}, \quad E_2 = E \setminus \{x_2\}, \quad E_{12} = E \setminus \{x_1, x_2\}.$$

These are sieves of E containing A ; moreover $E_1 \cup E_2 = E$ and $E_1 \cap E_2 = E_{12}$. The restriction of r to each E_T , for $T = 1, 2, 12$, still takes values in A and satisfies (5.1). Each complement $E_T \setminus A$ is smaller. Define

$$Y_T = X \amalg_{U_A} U_{E_T}, \quad j_T : X \longrightarrow Y_T.$$

By induction every $N_n j_T$ is a weak homotopy equivalence.

Corollary 4.5(i), applied to the sieves E_1, E_2 containing A and covering E , identifies the canonical maps $Y_i \rightarrow Y$ as full sieve inclusions. Their full intersection is canonically Y_{12} , and they cover the objects of Y . These identifications hold for the given arbitrary attaching functor a .

Lemma 4.6 now identifies the bottom pushout in the following diagram with $N_n Y$:

$$\begin{array}{ccccc} N_n X & \xlongequal{\quad} & N_n X & \xlongequal{\quad} & N_n X \\ N_n j_1 \downarrow & & \downarrow N_n j_{12} & & \downarrow N_n j_2 \\ N_n Y_1 & \longleftarrow & N_n Y_{12} & \longrightarrow & N_n Y_2. \end{array} \tag{5.4}$$

The vertical maps are the canonical maps induced by X , so this is a strictly commuting map of spans. All three are weak equivalences by induction, and the indicated legs in both rows are monomorphisms. Lemma 2.5 proves that the induced map on pushouts is a weak equivalence. The top pushout is $N_n X$, and its map to the bottom pushout is precisely $N_n j$, by its restrictions to the two pieces. This completes the finite induction. In particular, no compatibility between the local cylinders on different pieces has been used.

Now let E be arbitrary, and write $q = ir : E \rightarrow E$. Since $q^2 = q$, the collection $\mathcal{F}$ of finite subsets $F \subset E$ with $q(F) \subset F$ is nonempty and filtered under inclusion: it contains $\emptyset$, and finite unions remain stable. Every finite subset T is contained in the finite stable subset $T \cup q(T)$. Give each F its full induced order and put $A_F = A \cap F$. Then A_F is a sieve in F , and stability gives $r(F) \subset A_F$. The restriction $r_F : F \rightarrow A_F$ is a right adjoint retraction, since (5.1) restricts to these full subposets. Notice that F itself need not be a sieve in E .

Use the restriction of a to define

$$Y_F = X \amalg_{U_{A_F}} U_F, \quad j_F : X \longrightarrow Y_F.$$

Every j_F is a Thomason equivalence by the finite case. The inclusions $F \subset F'$ give compatible transition functors $Y_F \rightarrow Y_{F'}$ under the same fixed X . Every ordinary nerve simplex has finite vertex support, and therefore

$$NE = \operatorname{colim}_{F \in \mathcal{F}} NF, \quad NA = \operatorname{colim}_{F \in \mathcal{F}} NA_F.$$

Applying the colimit-preserving left adjoint c_n and then commuting colimits gives

$$\operatorname{colim}_{F \in \mathcal{F}} Y_F \cong X \amalg_{U_A} U_E = Y. \quad (5.5)$$

For clarity, the fixed copy of X in this formula is essential and legitimate: a compatible family of maps $Y_F \rightarrow Z$ consists of one map $X \rightarrow Z$, since $\mathcal{F}$ is nonempty and connected, together with compatible maps $U_F \rightarrow Z$ agreeing on each U_{A_F} . These are exactly maps from the pushout on the right. Under this identification the colimit of the arrows j_F is j .

By Lemma 2.4, N_n preserves these filtered colimits and simplicial weak homotopy equivalences are closed under filtered colimits. Hence $N_n j$ is a weak homotopy equivalence. No injectivity of the transition functors $Y_F \rightarrow Y_{F'}$ is required. The same proof applies directly when $n = \infty$. $\square$

The following unit comparison is known; see [5, Corollary 10.11], [9, Theorem 6.9] and [3, Proposition 9.2.3]. We include an independent proof using the same prism and sieve-cover method.

Proposition 5.3 (The unit on every poset). *For every small poset S , the adjunction unit*

$$\eta_S : NS \longrightarrow N_n U_S$$

is a weak homotopy equivalence. This assertion is natural in S and holds for every finite $n \geq 1$ and for $n = \infty$.

Proof. First take S finite and induct on its cardinality. If S is empty, both simplicial sets are empty and the unit is an isomorphism. If S has a greatest element x , the pointwise inequality $\operatorname{id}_S \leq \operatorname{const}_x$ contracts NS . Lemma 3.1, with old sieve $\emptyset$, gives a right cylinder from id_{U_S} to $c_n N(\operatorname{const}_x)$. The latter functor factors through $U_{\{x\}} = \mathcal{O}_0$, the terminal strict n -category. Lemma 2.3 therefore contracts $N_n U_S$. Both realizations are nonempty and contractible, so η_S is a weak homotopy equivalence.

In the remaining case choose two distinct maximal elements x_1, x_2 of S and put $S_i = S \setminus \{x_i\}$ and $S_{12} = S \setminus \{x_1, x_2\}$. These are sieves, the two single deletions cover S , and all three posets are smaller. Their ordinary nerves cover NS : a simplex belongs to a sieve containing its last vertex. By Lemmas 4.3 and 4.6, their $U_\cdot$ images give a full sieve cover of U_S and the analogous cover of its Street nerve. The joint zero label identifies the intersection of the two full sieves with $U_{S_{12}}$. Naturality of the unit gives a commuting map of spans

$$\begin{array}{ccccc} NS_1 & \longleftarrow & NS_{12} & \longleftarrow & NS_2 \\ \eta_{S_1} \downarrow & & \downarrow \eta_{S_{12}} & & \downarrow \eta_{S_2} \\ N_n U_{S_1} & \longleftarrow & N_n U_{S_{12}} & \longleftarrow & N_n U_{S_2}. \end{array}$$

The vertical maps are weak equivalences by induction; all horizontal legs are monomorphisms. The induced map on pushouts is a weak equivalence by Lemma 2.5. Under the cover identifications it is exactly η_S : its restriction to either NS_i agrees with that unit by naturality. This proves the finite case.

Finally every poset is the filtered union of its finite full subposets. Ordinary nerve preserves this union because each simplex has finite support, c_n preserves its colimit, and N_n preserves the resulting filtered colimit by Lemma 2.4. Naturality identifies η_S with the filtered colimit of the units on those finite subposets. Filtered-colimit closure of simplicial weak homotopy equivalences then proves the assertion. This proof uses only the prism, cover, and simplicial gluing lemmas, and in particular precedes and is independent of the model-structure transfer. $\square$

6 Comparison for subdivided attachments

We now compare an ordinary pushout of Street nerves with the Street nerve of an actual attachment. All attaching functors in this section are arbitrary. In particular, they may identify old objects and cells. The comparison will supply both the acyclicity of horn attachments and the preservation of weak equivalences under boundary attachments.

We first treat a sieve $A \subset B$ whose inclusion into its upward closure admits a right adjoint retraction. Couniversality controls the attachment over that upward closure; a cosieve cover then adds the rest of B . This decomposition follows the proof of [17, Proposition 4.3]; Theorem 5.2 supplies its homotopy input in strict higher dimensions. After proving this general comparison, we identify the twice-subdivided boundaries and horns as poset pairs with these properties.

Proposition 6.1 (Attachment comparison). *Let $A \subset B$ be a sieve of small posets. Let Z be its upward closure, and suppose that $A \hookrightarrow Z$ admits a right adjoint retraction. For every functor $a : U_A \rightarrow X$ in $n\text{Cat}$, put*

$$V_B(X) = X \amalg_{U_A} U_B, \quad P_B(X) = N_n X \amalg_{N_n U_A} N_n U_B.$$

The canonical map

$$\gamma_{B,X} : P_B(X) \longrightarrow N_n V_B(X) \tag{6.1}$$

is a simplicial weak equivalence. The map $N_n U_A \rightarrow N_n U_B$ is a monomorphism. The comparison is natural in every functor $f : X \rightarrow X'$ under U_A .

Proof. First set

$$V_Z(X) = X \amalg_{U_A} U_Z, \quad P_Z(X) = N_n X \amalg_{N_n U_A} N_n U_Z.$$

Theorem 5.2, applied to $A \hookrightarrow Z$, says that $X \rightarrow V_Z(X)$ belongs to W_n . Applying it with old target U_A also says that $U_A \rightarrow U_Z$ belongs to W_n . Since A is a sieve in Z , Lemma 4.3 identifies $U_A \rightarrow U_Z$ with an actual full inclusion. Its Street nerve is therefore a monomorphism and a weak equivalence, hence a trivial cofibration of simplicial sets. Its pushout $N_n X \rightarrow P_Z(X)$ is a weak equivalence. The canonical triangle

$$\begin{array}{ccc} N_n X & \longrightarrow & P_Z(X) \\ & \searrow & \downarrow \gamma_{Z,X} \\ & & N_n V_Z(X) \end{array}$$

then shows by two-out-of-three that $\gamma_{Z,X}$ is a weak equivalence.

To pass from Z to B , write

$$C = B \setminus A, \quad T = Z \setminus A = Z \cap C.$$

Both Z and C are cosieves in B , and they cover B : an element outside C belongs to $A \subset Z$. Their intersection is T . Lemma 4.3 therefore gives cosieves $U_Z, U_C \subset U_B$ with intersection U_T . Apply

Corollary 4.5(ii) with $P = B$, $T_1 = Z$, and $T_2 = C$. Its remaining hypotheses hold because $A \subset Z$ and $A \cap C = \emptyset$. The canonical maps identify

$$V_Z(X), \quad U_C \subset V_B(X), \quad V_Z(X) \cap U_C = U_T, \quad (6.2)$$

as actual full cosieves and their full intersection. These two cosieves cover all objects of $V_B(X)$. The corollary applies to the given arbitrary functor a , including its identifications of old objects and cells.

Apply Lemma 4.6 to the two cosieve covers just obtained. It gives ordinary simplicial pushouts

$$N_n U_B = N_n U_Z \amalg_{N_n U_T} N_n U_C, \quad (6.3)$$

$$N_n V_B(X) = N_n V_Z(X) \amalg_{N_n U_T} N_n U_C. \quad (6.4)$$

All maps from the intersections to the pieces in these covers are monomorphisms. Reassociating the pushout in the definition of $P_B(X)$ using Equation (6.3) gives

$$P_B(X) = P_Z(X) \amalg_{N_n U_T} N_n U_C.$$

Under these canonical identifications, Equation (6.1) is exactly the map induced by $\gamma_{Z,X}$ and the identity of $N_n U_C$. In particular the square

$$\begin{array}{ccc} P_Z(X) & \hookrightarrow & P_B(X) \\ \gamma_{Z,X} \downarrow & & \downarrow \gamma_{B,X} \\ N_n V_Z(X) & \hookrightarrow & N_n V_B(X) \end{array}$$

is a pushout: its lower right corner is also $N_n V_Z(X) \amalg_{P_Z(X)} P_B(X)$. The top horizontal map is the cobase change of the specific monomorphism

$$N_n U_T \hookrightarrow N_n U_C. \quad (6.5)$$

Left properness of simplicial sets, as recorded in Lemma 2.5, now makes $\gamma_{B,X}$ a weak equivalence. Neither the map $N_n U_A \rightarrow N_n X$ nor the map $N_n U_T \rightarrow P_Z(X)$ needs to be a monomorphism.

The remaining assertion that $N_n U_A \rightarrow N_n U_B$ is monic follows directly from the sieve identification in Lemma 4.3. Finally, $\gamma_{B,X}$ is defined by the two canonical maps from $N_n X$ and $N_n U_B$. For $f : X \rightarrow X'$ under U_A these maps commute with $N_n f$ and with the induced pushout functor $V_B(X) \rightarrow V_B(X')$. The pushout universal property therefore gives the strict naturality square

$$\begin{array}{ccc} P_B(X) & \xrightarrow{\gamma_{B,X}} & N_n V_B(X) \\ \downarrow & & \downarrow \\ P_B(X') & \xrightarrow{\gamma_{B,X'}} & N_n V_B(X'). \end{array} \quad (6.6)$$

The comparison is thus canonical and natural, while the local contractions used in Theorem 5.2 establish that it is a weak equivalence. $\square$

To apply the comparison to the generating maps, we now describe their second subdivisions. We use the classical face-chain description and intersection retraction of [17, Proposition 4.2]; see also [4, Lemmas 4.8 and 4.10]. The proof below records the precise poset pair for each boundary and horn, including the empty case.

Lemma 6.2 (The subdivision posets). *Let $d : K \hookrightarrow \Delta[m]$ be a boundary inclusion, with $m \geq 0$, or a horn inclusion $\Lambda^k[m] \hookrightarrow \Delta[m]$, with $m \geq 1$ and $0 \leq k \leq m$. There are finite posets $A \subset B$ and a canonical identification*

$$\mathrm{Sd}^2 d = (NA \hookrightarrow NB).$$

Here A is a sieve. Its upward closure Z in B is a cosieve, and $A \hookrightarrow Z$ has a right adjoint retraction.

Proof. Put $M = \{0, \dots, m\}$ and let P be the poset of its nonempty subsets, ordered by inclusion. The subposet P_K of faces belonging to K is

$$P_K = \begin{cases} \{F : \emptyset \neq F \subsetneq M\}, & K = \partial\Delta[m], \\ \{F : \emptyset \neq F \subseteq M, M \setminus \{k\} \not\subseteq F\}, & K = \Lambda^k[m]. \end{cases}$$

Indeed, a nonempty face belongs to the horn precisely when it is contained in $M \setminus \{j\}$ for some $j \neq k$. Thus in the horn case P_K is obtained from P by removing M and $M \setminus \{k\}$. In both cases P_K is downward closed in P .

The barycentric subdivision of an ordered simplicial complex has a vertex for every nonempty face and a nondegenerate simplex for every strict chain of such faces. Consequently

$$\mathrm{Sd}\Delta[m] = NP, \quad \mathrm{Sd}K = NP_K.$$

One can see these identifications directly on a simplex: the vertices of its subdivision are its nonempty subsets, and its simplices are chains of those subsets. The identifications agree on every face, so they glue over the faces of K . The nondegenerate simplices of NP are themselves the nonempty finite chains in P , with a face obtained by deleting entries. Applying the same description a second time gives

$$\begin{aligned} B &= \{\sigma : \sigma \text{ is a nonempty chain in } P\}, \\ A &= \{\sigma \in B : \sigma \subseteq P_K\}, \end{aligned} \quad \sigma \leq \tau \iff \sigma \subseteq \tau.$$

This proves the claimed identifications as simplicial sets, including degeneracies, and identifies their inclusion.

Every nonempty subchain of a chain in A again belongs to A , so A is a sieve in B . Its upward closure is exactly

$$Z = \{\sigma \in B : \sigma \cap P_K \neq \emptyset\}.$$

For the nontrivial implication in this equality, a face $F \in \sigma \cap P_K$ gives $\{F\} \in A$ below σ . Define

$$r : Z \longrightarrow A, \quad r(\sigma) = \sigma \cap P_K.$$

This is a nonempty chain, is monotone under inclusion, and fixes A . For $\alpha \in A$ and $\sigma \in Z$ one has

$$\alpha \subseteq \sigma \iff \alpha \subseteq \sigma \cap P_K = r(\sigma).$$

This is precisely the right adjunction to $A \hookrightarrow Z$.

When $m = 0$ and $K = \partial\Delta[0]$, the poset B is a singleton and $A = Z = \emptyset$; the asserted adjunction is the identity of the empty poset. For a one-dimensional horn, $P_K = \{\{k\}\}$, so A is a singleton and Z consists of the chains containing the face $\{k\}$. These cases are included in the formulas above. There are no zero-dimensional horn generators. $\square$

Remark 6.3. For the zero-dimensional boundary, $A = Z = T = \emptyset$ and $B = C$ is a singleton. Thus $V_Z(X) = X$ and $\gamma_{Z,X} = \text{id}$; the two cosieves of $V_B(X) = X \amalg U_B$ are disjoint. The comparison in Proposition 6.1 is then the isomorphism

$$N_n X \amalg \Delta[0] \cong N_n(X \amalg U_B).$$

Corollary 6.4 (Horn attachments are acyclic). *For every horn inclusion $d \in J$, every pushout of $L_n d$ belongs to W_n .*

Proof. Use the pair $A \subset B$ from Lemma 6.2, so that $L_n d$ is $U_A \rightarrow U_B$. The inclusion $NA \rightarrow NB$ is a weak equivalence: the horn inclusion is a simplicial weak equivalence, and subdivision preserves weak equivalences. The latter follows by naturality and two-out-of-three from the objectwise weak equivalence $\text{Sd} \rightarrow \text{id}$ of [4, Theorem 4.5]. Naturality of the adjunction unit gives a square

$$\begin{array}{ccc} NA & \longrightarrow & NB \\ \downarrow & & \downarrow \\ N_n U_A & \longrightarrow & N_n U_B. \end{array}$$

The vertical maps are weak equivalences by Proposition 5.3. Two-out-of-three makes the lower map a weak equivalence, and it is monic by Proposition 6.1. Thus it is a simplicial trivial cofibration. For any attaching functor $U_A \rightarrow X$, its pushout $N_n X \rightarrow P_B(X)$ is a weak equivalence. The composite

$$N_n X \longrightarrow P_B(X) \xrightarrow{\gamma_{B,X}} N_n V_B(X)$$

is the nerve of the attachment $X \rightarrow V_B(X)$, and both factors are weak equivalences. This is exactly the assertion that the attachment belongs to W_n . $\square$

Corollary 6.5 (Invariance under attachment). *Let $d \in I \cup J$, write $L_n d$ as $U_A \rightarrow U_B$ using Lemma 6.2, and let $f : X \rightarrow X'$ belong to W_n . For every $a : U_A \rightarrow X$, the induced functor*

$$X \amalg_{U_A} U_B \longrightarrow X' \amalg_{U_A} U_B,$$

where the second attaching functor is fa , belongs to W_n .

Proof. The map $N_n X \rightarrow P_B(X)$ is a monomorphism, being the cobase change of $N_n U_A \hookrightarrow N_n U_B$. There is a canonical pushout identification

$$P_B(X') = N_n X' \amalg_{N_n X} P_B(X).$$

Left properness of simplicial sets therefore makes $P_B(X) \rightarrow P_B(X')$ a weak equivalence, since $N_n f$ is one. In the naturality square of Equation (6.6), both horizontal maps are weak equivalences by Proposition 6.1. Two-out-of-three implies that its right vertical map is a weak equivalence. This is the nerve of the asserted functor. $\square$

7 The model structure and the Quillen equivalence

Fix $n \in \{1, 2, \dots, \infty\}$. Corollary 6.4 supplies the acyclic horn attachments needed for the model axioms; Corollary 6.5 supplies the invariance under boundary attachment needed for left properness. We extend these statements through transfinite compositions, then prove the Quillen equivalence by checking the unit on arbitrary simplicial sets. Write

$$L_n = c_n \text{Sd}^2, \quad R_n = \text{Ex}^2 N_n, \quad L_n \dashv R_n.$$

We use the Kan–Quillen model structure on $\mathbf{sSet}$, with weak equivalences $W_{\mathbf{KQ}}$, fibrations $\mathbf{Fib}_{\mathbf{KQ}}$, and generating sets

$$\begin{aligned} I &= \{\partial\Delta[m] \hookrightarrow \Delta[m] \mid m \geq 0\}, \\ J &= \{\Lambda^k[m] \hookrightarrow \Delta[m] \mid m \geq 1, 0 \leq k \leq m\}. \end{aligned}$$

In particular, $\partial\Delta[0] = \emptyset$.

We first record the classical simplicial facts needed to identify the weak equivalences. The last-vertex transformation $\ell : \mathbf{Sd} \rightarrow \mathbf{id}$ is an objectwise weak equivalence [4, Theorem 4.5]. Its mate $j : \mathbf{id} \rightarrow \mathbf{Ex}$ is also an objectwise weak equivalence [10, Theorem III.4.6]. The latter assertion is a theorem about subdivision and extension, and is not a formal consequence of being a mate. Both statements apply to every simplicial set independently of the model structure being constructed. For a map $f : K \rightarrow K'$, naturality gives

$$j_{K'}f = (\mathbf{Ex}f)j_K.$$

Two-out-of-three therefore proves that $\mathbf{Ex}$ preserves and reflects weak equivalences. Similarly, naturality of ℓ shows that $\mathbf{Sd}$ preserves and reflects weak equivalences. In particular,

$$\begin{aligned} W_n &= N_n^{-1}(W_{\mathbf{KQ}}) = R_n^{-1}(W_{\mathbf{KQ}}), \\ F_n &= R_n^{-1}(\mathbf{Fib}_{\mathbf{KQ}}), \quad C_n = {}^\perp(F_n \cap W_n). \end{aligned} \tag{7.1}$$

Here ${}^\perp$ denotes the left lifting property. The class W_n has two-out-of-three and is closed under retracts. By Lemma 2.4, it is also closed under filtered colimits of arrows, without any injectivity requirement on their transition maps.

For a set $\mathcal{S}$ of maps, write $\mathcal{S}\text{-inj}$ for the maps with the right lifting property against $\mathcal{S}$, and $\mathcal{S}\text{-cof} = {}^\perp(\mathcal{S}\text{-inj})$. A relative $\mathcal{S}$ -cell map is a transfinite composite of pushouts of coproducts of members of $\mathcal{S}$.

Lemma 7.1 (Transfinite horn attachments). *Every relative $L_n J$ -cell map belongs to W_n .*

Proof. Corollary 6.4 proves the assertion for one attachment, with an arbitrary attaching target. A set of simultaneous attachments may be well ordered and attached one at a time: at each stage use the given map from its source into the original target, followed by the map to the current stage. The universal property of the resulting colimit is the universal property of the pushout of the coproduct. Thus it suffices to consider a continuous ordinal sequence $(X_\alpha)_{\alpha \leq \lambda}$ in which each successor map is a single such attachment.

By transfinite induction, $X_0 \rightarrow X_\alpha$ belongs to W_n for every α . The assertion at zero is the identity, and the successor step is composition with the weak equivalence supplied by Corollary 6.4. If β is a nonzero limit ordinal, the ordinal of stages $\alpha < \beta$ is filtered. Apply Lemma 2.4 to the diagram of arrows $X_0 \rightarrow X_\alpha$. Its colimit is $X_0 \rightarrow \operatorname{colim}_{\alpha < \beta} X_\alpha = X_\beta$, because the colimit of the constant source diagram is X_0 . This proves the limit step and the lemma. $\square$

Proposition 7.2 (The transferred model structure). *The classes (C_n, W_n, F_n) in (7.1) form a combinatorial model structure on $n\mathbf{Cat}$, cofibrantly generated by $(L_n I, L_n J)$.*

Proof. The underlying category is locally presentable, and hence complete and cocomplete, by Lemma 2.4. We describe the small object argument to make clear that the factorizations have no restriction on their sources. For either set $\mathcal{S} = L_n I$ or $\mathcal{S} = L_n J$, choose an infinite regular cardinal

κ such that every domain in $\mathcal{S}$ is κ -presentable. Such a common bound exists because the category is locally presentable and $\mathcal{S}$ is a set.

Given any map $X \rightarrow Y$, start with $X_0 = X$. At stage $\alpha + 1$ attach to X_α one copy of the codomain of each member of $\mathcal{S}$ for every commutative square from that member to $X_\alpha \rightarrow Y$. This is a set of squares, by local smallness. Use colimits at limit stages. The resulting factorization is

$$X \xrightarrow{i} X_\kappa \xrightarrow{p} Y.$$

Here i is a relative $\mathcal{S}$ -cell map and $p \in \mathcal{S}\text{-inj}$. Indeed a map from a generating domain to X_κ factors through some X_α , since the ordinal κ is κ -filtered. Its lifting square against p was therefore attached at stage $\alpha + 1$, which supplies the required lift. The construction may be made functorial by using all squares at each stage.

Pushouts, transfinite composites, and retracts of maps with a fixed left lifting property retain that property. Consequently every relative $\mathcal{S}$ -cell map lies in $\mathcal{S}\text{-cof}$. Conversely, if $u : A \rightarrow B$ belongs to $\mathcal{S}\text{-cof}$, factor it as $A \xrightarrow{i} Z \xrightarrow{p} B$ by the construction above. The square with top map i , left map u , right map p , and bottom map id_B admits a diagonal $s : B \rightarrow Z$. The equations $su = i$ and $ps = \text{id}_B$ exhibit u as a retract of i in the arrow category, with the source fixed at A . Thus the construction gives the weak factorization system

$$(\mathcal{S}\text{-cof}, \mathcal{S}\text{-inj}),$$

and its left class consists exactly of retracts of relative $\mathcal{S}$ -cell maps.

The adjunction identifies lifting squares and their fillers. The Kan–Quillen lifting characterizations therefore give exactly

$$(L_n I)\text{-inj} = R_n^{-1}(I\text{-inj}) = F_n \cap W_n, \quad (7.2)$$

$$(L_n J)\text{-inj} = R_n^{-1}(J\text{-inj}) = F_n. \quad (7.3)$$

In the first equality of classes on the right of (7.2), we used the equality of the two definitions of W_n proved above. The first weak factorization system is consequently $(C_n, F_n \cap W_n)$.

Let $\mathcal{D} = (L_n J)\text{-cof}$. As $F_n \cap W_n \subseteq F_n$, one has $\mathcal{D} \subseteq C_n$. Lemma 7.1 and retract closure give $\mathcal{D} \subseteq W_n$ as well. To prove the reverse inclusion, take $u : A \rightarrow B$ in $C_n \cap W_n$ and factor it by the second small object argument as $A \xrightarrow{i} Z \xrightarrow{p} B$, with $i \in \mathcal{D}$ and $p \in F_n$. Since i and $u = pi$ belong to W_n , two-out-of-three gives $p \in F_n \cap W_n$. The lifting property of $u \in C_n$ now supplies $s : B \rightarrow Z$ with $su = i$ and $ps = \text{id}_B$. The same fixed-source retract diagram puts u in $\mathcal{D}$. We have proved

$$\mathcal{D} = C_n \cap W_n.$$

Thus the second weak factorization system is $(C_n \cap W_n, F_n)$. Together with two-out-of-three and retract closure for W_n , these are all the model axioms. Local presentability and the displayed generating sets make the model structure combinatorial. $\square$

Proposition 7.3 (Properness). *The model structure of Proposition 7.2 is left and right proper.*

Proof. For right properness, let $p : Y \rightarrow Z$ belong to F_n and let $w : Z' \rightarrow Z$ belong to W_n . The right adjoint R_n carries the pullback $Y \times_Z Z'$ to the corresponding simplicial pullback. Its map to $R_n Y$ is a weak equivalence by right properness of the Kan–Quillen model structure, since $R_n p$ is a Kan fibration and $R_n w$ is a weak equivalence. The definition of W_n then gives $Y \times_Z Z' \rightarrow Y$ in W_n .

For left properness, first let $i : X \rightarrow Y$ be a relative $L_n I$ -cell map and let $w : X \rightarrow X'$ belong to W_n . As in Lemma 7.1, write i as a continuous sequence of individual boundary attachments X_α

starting at X . Put $X'_\alpha = X' \amalg_X X_\alpha$. We prove by induction that $X_\alpha \rightarrow X'_\alpha$ belongs to W_n . This is w at stage zero. At a successor stage, the second attachment has the first attaching map followed by $X_\alpha \rightarrow X'_\alpha$, so Corollary 6.5 proves the induction step. At a nonzero limit ordinal, both sequences are their colimits, and Lemma 2.4 gives the assertion. Hence $Y \rightarrow X' \amalg_X Y$ belongs to W_n .

For a general $i : X \rightarrow Y$ in C_n , the first small object factorization and its lifting retract, used in Proposition 7.2, express i as a retract of a relative $L_n I$ -cell map $j : X \rightarrow Z$ with the same source. Explicitly there are $s : Y \rightarrow Z$ and $p : Z \rightarrow Y$ with $si = j$, $pj = i$, and $ps = \text{id}_Y$. Cobase change along w gives the corresponding maps s' and p' between $X' \amalg_X Y$ and $X' \amalg_X Z$, with $p's' = \text{id}$. Thus the arrow $Y \rightarrow X' \amalg_X Y$ is a retract of $Z \rightarrow X' \amalg_X Z$. The latter is in W_n by the preceding paragraph, and so is the former. Both properness arguments apply to arbitrary objects; no cofibrancy or fibrancy condition on the base objects was used. $\square$

Let $\eta_T : T \rightarrow N_n c_n T$ denote the unit of $c_n \dashv N_n$. The poset unit of Proposition 5.3 is the case $T = NE$. The next proposition extends the required comparison to all simplicial sets after two subdivisions.

Proposition 7.4 (The unit after two subdivisions). *For every simplicial set K , the canonical map*

$$\theta_K = \eta_{\text{Sd}^2 K} : \text{Sd}^2 K \longrightarrow N_n L_n K \quad (7.4)$$

is a weak equivalence.

Proof. For $K = \Delta[m]$ and $K = \partial\Delta[m]$, Lemma 6.2 identifies $\text{Sd}^2 K$ with the ordinary nerve of a poset. The assertion is therefore Proposition 5.3. It also holds for the empty simplicial set: $c_n \emptyset$ is the empty strict category, whose Street nerve is empty.

Every simplicial set is obtained from the empty set by a continuous transfinite sequence of boundary attachments. One construction orders its nondegenerate simplices by dimension and well orders those in each dimension. The boundary of a simplex then maps into the stage already constructed. That attaching map may identify vertices or faces, or send faces to degenerate simplices. The construction still recovers K , by the unique expression of each simplex as a degeneracy of a nondegenerate simplex.

Consider one successor step

$$K_+ = K_0 \amalg_{\partial\Delta[m]} \Delta[m]$$

and assume the assertion for K_0 . Write

$$V = \text{Sd}^2 K_0, \quad A = \text{Sd}^2 \partial\Delta[m], \quad B = \text{Sd}^2 \Delta[m].$$

Here A, B, V are simplicial sets; the representing posets for A, B will be denoted by E, F below. Since subdivision is a left adjoint, $\text{Sd}^2 K_+ = V \amalg_A B$. Naturality of η gives a map of spans and hence the map

$$q : V \amalg_A B \longrightarrow N_n c_n V \amalg_{N_n c_n A} N_n c_n B. \quad (7.5)$$

The three maps on the objects of these spans are weak equivalences: on V by induction and on A, B by the simplex and boundary cases. The leg $A \rightarrow B$ is a simplicial monomorphism. Writing $A = NE$ and $B = NF$ as in Lemma 6.2, the inclusion $E \subseteq F$ is a sieve; Lemma 4.3 makes $c_n A \rightarrow c_n B$ a sieve inclusion. Consequently $N_n c_n A \rightarrow N_n c_n B$ is also a monomorphism. Lemma 2.5, applied with these two specified monomorphic legs, proves that q is a weak equivalence. It requires no monomorphism condition on $A \rightarrow V$.

Apply Proposition 6.1 to the arbitrary attaching functor $c_n A \rightarrow c_n V$. It says that

$$t : N_n c_n V \amalg_{N_n c_n A} N_n c_n B \longrightarrow N_n (c_n V \amalg_{c_n A} c_n B) \quad (7.6)$$

is a weak equivalence. Because c_n preserves colimits, the target is $N_n c_n(V \amalg_A B)$. The composite tq is exactly $\eta_{V \amalg_A B}$: on each of the two pushout legs V and B this follows from naturality of η , and equality on those legs determines the map from the pushout. Thus the actual canonical map θ_{K_+} is a weak equivalence. This also explains why noninjective boundary maps cause no change to the argument.

At a nonzero limit ordinal, Sd^2 and L_n preserve the colimit as left adjoints, while N_n preserves the filtered colimit by Lemma 2.4. Therefore the unit at that stage is the filtered colimit of the previous units; it is a weak equivalence by the same lemma. Transfinite induction proves (7.4) for every K . The poset descriptions used in this proof concern the simplex and boundary pieces only. $\square$

Theorem 7.5 (Quillen equivalence). *The adjunction*

$$L_n : \text{sSet}_{\text{KQ}} \rightleftarrows (n\text{Cat}, C_n, W_n, F_n) : R_n$$

is a Quillen equivalence.

Proof. Equations (7.2) and (7.3) show that R_n preserves fibrations and trivial fibrations, so this is a Quillen adjunction.

Set $S = \text{Sd}$ and $T = \text{Ex}$, and let $a_K : K \rightarrow TSK$ be the unit of $S \dashv T$. The mate identity is

$$j_K = T(\ell_K)a_K : K \longrightarrow TK.$$

The independently cited theorem above makes j_K a weak equivalence. Since ℓ_K is a weak equivalence and T preserves weak equivalences, $T(\ell_K) : TSK \rightarrow TK$ is also one. Hence a_K is a weak equivalence for every simplicial set K , by two-out-of-three. In particular, $a_{SK} : SK \rightarrow TS^2K$ is a weak equivalence. The unit of $S^2 \dashv T^2$ is

$$b_K = T(a_{SK})a_K : K \longrightarrow T^2S^2K, \quad T(a_{SK}) : TSK \longrightarrow T^2S^2K.$$

Thus b_K is a weak equivalence because T preserves weak equivalences. Composition is written with the rightmost map applied first.

The unit of the composite adjunction $L_n \dashv R_n$ is

$$u_K = T^2(\theta_K)b_K : K \longrightarrow T^2N_n c_n S^2K = R_n L_n K.$$

Proposition 7.4, preservation of weak equivalences by T , and the preceding calculation show that u_K is a weak equivalence for every K . For any strict functor $f : L_n K \rightarrow Y$, its simplicial adjunct is

$$f^\sharp = R_n(f)u_K : K \longrightarrow R_n Y.$$

Using the definition of W_n and two-out-of-three now gives

$$f \in W_n \iff R_n(f) \in W_{\text{KQ}} \iff f^\sharp \in W_{\text{KQ}}.$$

This equivalence holds for every K and Y . In particular it holds for cofibrant K and fibrant Y , which is the Quillen-equivalence criterion. Together with Propositions 7.2 and 7.3, this proves all the asserted model-categorical conclusions for each finite $n \geq 1$ and for $n = \infty$. $\square$

A Presentations and local finite presentability

We give a direct proof of the presentability assertion used in Lemma 2.4. The point requiring care is that an equation between boundary cells can create new composable pairs. Presentations must therefore generate defined terms and equations together; taking an ordinary quotient of the cells already present would not suffice. All sets and constructions below lie in the fixed universe.

A.1 Operations and their domains

Fix $n \in \{1, 2, \dots, \infty\}$, with ∞ denoting ω . Use one sort C_d for each $0 \leq d \leq n$, or for each nonnegative integer d when $n = \infty$. The total operations are

$$s_d, t_d : C_d \longrightarrow C_{d-1} \quad (d > 0), \quad e_d : C_d \longrightarrow C_{d+1} \quad (d < n).$$

For $0 \leq k < d$, write s_k^d, t_k^d for the iterated source and target, and e_k^d for the iterated identity from sort k to sort d . Each is a finite term. There is a binary partial operation $*_k^d$ on C_d whose domain is exactly

$$s_k^d(x) = t_k^d(y). \tag{A.1}$$

Thus $x *_k^d y$ means x after y . Superscripts will be omitted when the sort is clear.

For clarity, the equations imposed are the usual strict globular category equations. Globularity and the identity boundary equations are

$$s_{d-1}s_d = s_{d-1}t_d, \quad t_{d-1}s_d = t_{d-1}t_d, \quad s_{d+1}e_d = \text{id} = t_{d+1}e_d.$$

The globularity equations apply for $2 \leq d \leq n$, and the identity boundary equations for $0 \leq d < n$; when $n = \infty$, there is no upper bound on these integer indices. For a pair satisfying (A.1), the composition boundary equations are

$$\begin{aligned} s_d(x *_k y) &= s_d(y), & t_d(x *_k y) &= t_d(x), \\ s_d(x *_k y) &= s_d(x) *_k s_d(y), & t_d(x *_k y) &= t_d(x) *_k t_d(y) \quad (k < d-1). \end{aligned}$$

For each k , associativity is imposed on triples with $s_k(x) = t_k(y)$ and $s_k(y) = t_k(z)$. The unit equations are

$$x *_k e_k^d(s_k^d x) = x, \quad e_k^d(t_k^d x) *_k x = x.$$

Identities preserve every lower composition: $e_d(x *_k^d y) = e_d(x) *_k^{d+1} e_d(y)$ whenever the pair is composable and $d < n$.

Interchange, for $k < l < d$, is

$$(x *_l y) *_k (z *_l w) = (x *_k z) *_l (y *_k w). \tag{A.2}$$

Its guard includes all four inner matching equations:

$$\begin{aligned} s_l^d(x) &= t_l^d(y), & s_l^d(z) &= t_l^d(w), \\ s_k^d(x) &= t_k^d(z), & s_k^d(y) &= t_k^d(w). \end{aligned} \tag{A.3}$$

These equations also imply the two outer matching conditions. For the left side, globularity gives

$$s_k^d(x *_l y) = s_k^d(x) = s_k^d(y), \quad t_k^d(z *_l w) = t_k^d(z) = t_k^d(w).$$

For the right side, the iterated boundary formulas give

$$\begin{aligned} s_l^d(x *_k z) &= s_l^d(x) *_k^l s_l^d(z), \\ t_l^d(y *_k w) &= t_l^d(y) *_k^l t_l^d(w). \end{aligned}$$

The lower-dimensional composites in these formulas are defined by the last two equations of (A.3) and globularity. The first two equations then make their values equal. Thus the guard describes an ordinary composable array, not an additional restriction on interchange. Definedness of the right-hand side alone is not used to infer the guard or definedness of the left-hand side.

These are a set of conditional equations with finite terms and finitely many premises. In the ω case the sorts and operation symbols are countable, but no operation or equation involves infinitely many arguments. The resulting models are precisely strict globular n -categories. For finite n , adjoining iterated identities above dimension n gives the convention used in the body of the paper.

A.2 Defined terms and presentations

A *sorted set of generators* G assigns a dimension to each generator. Let $T(G)$ be the set of all finite well-sorted raw terms in these generators and the operation symbols. A formal composite belongs to $T(G)$ even if its boundary equation is not yet known; it will represent a cell only after being declared defined.

Let R be a set of equations between raw terms of the same sort. An equation in R is a *strong equation*: its interpretation requires both terms to be defined and their values to agree. In particular, $t = t$ can be used to require that a particular term be defined.

Lemma A.1 (Strict-category presentations). *For every sorted set G and every set R of strong equations, there is a small strict n -category $P(G, R)$ such that, naturally in Y , strict functors $P(G, R) \rightarrow Y$ correspond to sorted assignments of the generators to cells of Y satisfying R .*

Proof. Generate simultaneously judgments $D(t)$ and $t = u$, meaning that t is defined and that t, u are defined with the same value. Use the following rules.

- (i) Every generator is defined, and each member of R is an equation.
- (ii) An equation $t = u$ implies $D(t)$ and $D(u)$. Definedness of an operation term implies definedness of all its arguments. Definedness of $x *_k y$ also implies its matching equation (A.1).
- (iii) Total operations applied to defined arguments give defined terms. Defined arguments satisfying (A.1) give a defined composite.
- (iv) Equality is symmetric and transitive. Reflexivity is the rule $D(t) \Rightarrow t = t$; it is not imposed on undefined raw terms.
- (v) Equal arguments transport definedness and equality through an operation. Precisely, from $t_i = u_i$ for all arguments and $D(f(t_1, \dots, t_r))$, infer $D(f(u_1, \dots, u_r))$ and equality of these two operation terms.
- (vi) Impose every ground instance of the strict category equations above when its arguments are defined and its stated matching conditions hold. The conclusion is equality of the two sides and therefore definedness of both. In particular, interchange retains all four conditions in (A.3).

Globularity, identity boundary and unit equations require just definedness of their variables. Composition boundary equations and preservation of composition by identities require (A.1); associativity requires its two matching equations. Thus every rule has finitely many premises. All rules are sound in a strict category, including the definedness conclusions in the last rule.

Start with the assumptions and repeatedly add the conclusions of all applicable rules. The union of the finite stages is the least closed set of judgments: any rule's finitely many premises occur together at some stage. In particular, every resulting judgment has a finite derivation. This construction involves a set of judgments, since the signature and the set of raw finite terms are sets.

Let $P(G, R)_d$ be the defined terms of sort d modulo derivable equality. The equality rules give an equivalence relation on this set. Total operations descend by rules (iii) and (v). For the partial operations, suppose the boundary classes of $[x]$ and $[y]$ match. Total boundary terms are defined, so equality of their classes means that $s_k(x) = t_k(y)$ is derivable. Rule (iii) then declares $x *_k y$ defined.

This construction does not depend on the representatives. Indeed, if $x = x'$ and $y = y'$ are derivable, total-operation congruence gives

$$s_k(x) = s_k(x'), \quad t_k(y) = t_k(y').$$

The matching equality for x, y therefore gives the matching equality for x', y' . Rule (iii) defines both composites, and rule (v) equates them. Conversely, rule (ii) ensures that every admitted composite has matching boundaries. The operation domain in the quotient is consequently exactly the set of matching pairs. Rule (vi) gives all the strict category laws there.

Now let an assignment $G \rightarrow Y$ satisfy R . Induction on derivations shows that each term declared defined evaluates in Y and each derived equation has equal values. For rule (v), equal arguments have identical boundary values, so existence of one partial operation value entails existence of the other. Evaluation therefore descends to a strict functor $P(G, R) \rightarrow Y$. It is unique by induction on terms, since definedness is hereditary to subterms and the images of the generators are fixed. Conversely, every strict functor gives such an assignment. This proves the universal property.

The constructed object is small. There is no inconsistency obstruction: the strict category with one cell in each sort satisfies every strong equation under the unique assignment. If G is empty then there are no raw ground terms, since all operations have positive arity. The construction gives the empty strict category in that case. $\square$

Remark A.2. The simultaneous construction is necessary even for ordinary categories. If the source and target of the free arrow are identified, its presentation must create the square of the arrow and all higher powers. These are not obtained by merely identifying members of the original set of cells. Rule (iii) creates them as soon as their boundary equalities become derivable.

A.3 Limits, colimits, and finite presentations

Proposition A.3 (Local finite presentability). *For every $1 \leq n \leq \infty$, the category $n\text{Cat}$ is locally small, complete, cocomplete, and locally finitely presentable. Filtered colimits are computed on cells. Every set of objects has a common infinite regular presentability bound.*

Proof. Colimits. For a small diagram $X : I \rightarrow n\text{Cat}$, use a generator $[i, a]$ for each cell a of each X_i . Impose all its operation-table equations:

$$\begin{aligned} s([i, a]) &= [i, s(a)], & t([i, a]) &= [i, t(a)], \\ e([i, a]) &= [i, e(a)], & [i, a] *_{\mathbf{k}} [i, b] &= [i, a *_{\mathbf{k}} b] \end{aligned}$$

where the last equation is included for every defined composite. For each arrow $f : i \rightarrow j$ also impose $[i, a] = [j, X(f)(a)]$. All these generators and equations form sets. By Lemma A.1, maps from this presentation to Y are exactly compatible families of strict functors $X_i \rightarrow Y$. It is therefore the colimit.

Limits and local smallness. In each dimension, take the set of compatible families in the diagram. Source, target and identity are coordinatewise. Two families have matching boundaries exactly when their coordinates do, so coordinatewise composition is defined on exactly the required domain and again gives a compatible family. All laws and the limit universal property hold coordinatewise. This includes the empty limit, with a singleton in each sort. Strict functors form a subset of the product of the sets of cell maps, so the category is locally small.

Filtered colimits and finite witnesses. For a filtered diagram, form the colimit of each set of cells. Finitely many representatives occur at a common stage. If a matching equation holds in the colimit, it holds at a later common stage, where the composite can be formed. A further common stage proves independence of choices. Each category law holds at a stage containing its finite tuple and its matching equations. This constructs the colimit with its universal property, without any injectivity assumption on transition maps.

We will also use the finite-term version of this argument. If finitely many assigned generator values make a finite collection of terms defined and satisfy specified equations in the colimit, these

facts hold at a common stage. Induct on the finitely many subterms: first lift the arguments, then the boundary equality needed for a partial composite, and then the composite. Finally lift the required equalities. Each step uses the equality criterion for filtered colimits of sets.

Finite presentations. If G and R are finite, a map $P(G, R) \rightarrow \operatorname{colim}_i X_i$ is an assignment of finitely many generator values satisfying finitely many strong equations. The preceding finite-term argument lifts that assignment and all the relations to a single stage. Lemma A.1 then gives a functor to that stage. If two stage functors become equal in the colimit, their values on the finite set G agree at a later common stage; their entire functors agree there by the presentation universal property. Thus

$$\operatorname{colim}_i \operatorname{Hom}(P(G, R), X_i) \cong \operatorname{Hom}(P(G, R), \operatorname{colim}_i X_i).$$

Hence every finite presentation is finitely presentable. This does not assert that it has finitely many cells.

A generating set of finite presentations. Every strict category X has its full cell presentation: one generator g_a for each cell and all operation-table equations. Evaluation identifies its presented category with X . Indeed it is onto, and induction on a defined term, using the operation-table equations, equates that term to the generator of its value. This also proves injectivity.

Write this presentation as (G_X, R_X) . Consider all finite pairs (G_0, R_0) with $G_0 \subseteq G_X$, $R_0 \subseteq R_X$, and with every generator occurring in R_0 contained in G_0 . Ordered by inclusion, these pairs form a nonempty filtered poset: the empty pair is allowed, and finite unions give upper bounds. Their presented categories have colimit $P(G_X, R_X)$. To verify this, a compatible family of maps from the finite presentations assigns a value to every generator and enforces every relation, since each appears in some finite pair. The presentation universal property gives exactly a map from $P(G_X, R_X)$.

The finite pairs need not be closed under all operations. Finite presentations themselves supply the required closures. There is a set of finite presentation types: use finite numbered generator sets with specified sorts and finite sets of raw term equations. Thus every object is a filtered colimit of objects from a set of finitely presentable objects. Together with cocompleteness, this is local finite presentability. The same reasoning applies to the countable signature in the ω case.

A common regular bound. More generally, if G and R have cardinality less than an infinite regular cardinal κ , the same proof makes $P(G, R)$ κ -presentable. In a κ -filtered diagram one lifts fewer than κ generator values and the finitely many subterms of each of fewer than κ relations. All stage, morphism and equality witnesses involved form a diagram of size less than κ , so a cocone places them at one stage. Equality of maps is detected by the fewer than κ generator equalities in the same way.

The full cell presentation of X has cardinality at most

$$\max \left\{ \aleph_0, \left| \prod_d X_d \right| \right\},$$

because the signature is countable and operations have arity at most two. Choose a larger infinite regular cardinal. The preceding argument makes X presentable at that bound. For a set of objects, bound the cardinalities of all their full presentations first and choose one larger infinite regular cardinal. This gives the claimed common bound. $\square$

B Relation to the weak-groupoid expansion method

The elementary-expansion theorem of [13] concerns models of a fixed Grothendieck coherator in the Ara–Henry convention. Its Theorem 1.1 asserts that, for a cellular model X , freely adjoining a

parallel m -cell and an $(m + 1)$ -cell joining it to a prescribed cell preserves components and based homotopy groups. Cellular inputs include transfinite cellular presentations; its Corollary 5.1 extends the assertion to cofibrant retracts and gives a left semi-model structure. These hypotheses differ from those of Theorem 5.2, whose attaching target is an arbitrary strict n -category.

The construction in [13, Sections 3–4] first reflects a model to one with unique fillers above a fixed dimension q , preserving all cells through q . It then descends in the expansion dimension. Within one step, higher expansions already covered by the induction control upper cellular extensions. Finite tree diagrams are compared with strings using compositions and inverses. Duplicating the top generators and adjoining higher joining cells promotes the resulting generator homotopies to a statement about the entire free model. Partial cylinders, stationary below the working dimension, are then completed along their upper boundaries. Finally, a homotopy-group test uses only finitely many dimensions and therefore passes out of the cutoff. Filtered-colimit closure also handles transfinite cellular presentations.

There is a formal strict analogue of the cutoff: retain the cells of a strict ω -category through dimension q , and above q put one cell over each parallel boundary. Higher identities and compositions are forced by their boundaries, and all higher laws follow from uniqueness. This construction preserves the q -cells, whereas the reflection τ_q used in this paper can identify them. The formal analogy supplies no comparison with Street weak equivalences. For example, the inclusion of the object 0 in the ordinary category $0 \longrightarrow 1 \longleftarrow 2$ is a Street weak equivalence, since the category has a terminal object. There is nevertheless no single arrow joining 0 and 2 in either direction. Thus the weak-groupoid test by exact boundaries and single joining cells cannot serve as a test for Street weak equivalences. Inverse traversal in the tree comparison also requires an independent replacement in the directed setting.

In the present directed setting, the corresponding role is played by the positive relative prism of Lemma 3.1. It is stationary on the old sieve, and its chain formula gives compatible strict maps on whole right Gray cylinders. Lemma 2.1 passes these cylinders to finite n . When their endpoints fix the old sieve, Lemma 3.2 glues them at arbitrary attaching targets. A greatest new element gives two local comparisons of this form; finite sieve covers glue their weak-equivalence conclusions. Finite subposets stable under the retraction and filtered colimits then remove the finiteness restriction. The resulting attachment comparison applies to subdivided simplicial boundaries and horns and yields both unrestricted weak factorization systems. No result of [13] is a premise of this proof.

References

- [1] Jiří Adámek and Jiří Rosický. *Locally Presentable and Accessible Categories*, volume 189 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, Cambridge, 1994. doi: 10.1017/CBO9780511600579. URL <https://doi.org/10.1017/CBO9780511600579>. For the corrected proof of Theorem 3.36, see Adámek, Hébert and Rosický (1999).
- [2] Jiří Adámek, Michel Hébert, and Jiří Rosický. On essentially algebraic theories and their generalizations. *Algebra Universalis*, 41(3):213–227, 1999. doi: 10.1007/s000120050111. URL <https://doi.org/10.1007/s000120050111>.
- [3] Dimitri Ara. Théorie de l’homotopie des ∞ -catégories strictes. Habilitation à diriger des recherches, 2023. URL <https://www.normalesup.org/~ara/files/hdr.pdf>. Author’s 168-page manuscript; Sections 5.2.2–5.2.6 and 9.2.5–9.2.6.
- [4] Dimitri Ara and Georges Maltsiniotis. Vers une structure de catégorie de modèles à la Thomason sur la catégorie des n -catégories strictes. *Advances in Mathematics*, 259:557–654, 2014. doi:

- 10.1016/j.aim.2014.03.013. URL <https://arxiv.org/abs/1305.5086v3>. Numbering follows arXiv:1305.5086v3, 18 April 2014.
- [5] Dimitri Ara and Georges Maltsiniotis. Le type d'homotopie de la ∞ -catégorie associée à un complexe simplicial, 2015. URL <https://arxiv.org/abs/1503.02720v1>. Version 1, 9 March 2015.
 - [6] Dimitri Ara and Georges Maltsiniotis. *Joint et tranches pour les ∞ -catégories strictes*. Number 165 in Mémoires de la Société Mathématique de France, nouvelle série. Société Mathématique de France, 2020. ISBN 978-2-85629-921-0. doi: 10.24033/msmf.473. URL <https://arxiv.org/abs/1607.00668v4>. Numbering follows arXiv:1607.00668v4, 24 September 2020.
 - [7] Denis-Charles Cisinski. La classe des morphismes de Dwyer n'est pas stable par rétractes. *Cahiers de Topologie et Géométrie Différentielle Catégoriques*, 40(3):227–231, 1999. URL https://www.numdam.org/item/CTGDC_1999__40_3_227_0/.
 - [8] Thomas M. Fiore and Simona Paoli. A Thomason model structure on the category of small n -fold categories. *Algebraic & Geometric Topology*, 10(4):1933–2008, 2010. doi: 10.2140/agt.2010.10.1933. URL <https://msp.org/agt/2010/10-4/agt-v10-n4-p04-p.pdf>.
 - [9] Andrea Gagna. Strict n -categories and augmented directed complexes model homotopy types. *Advances in Mathematics*, 331:542–564, 2018. doi: 10.1016/j.aim.2018.04.010. URL <https://arxiv.org/abs/1612.04450v2>. Numbering follows arXiv:1612.04450v2, 15 February 2018.
 - [10] Paul G. Goerss and John F. Jardine. *Simplicial Homotopy Theory*, volume 174 of *Progress in Mathematics*. Birkhäuser, Basel, 1999. doi: 10.1007/978-3-0348-8707-6. URL <https://doi.org/10.1007/978-3-0348-8707-6>.
 - [11] Léonard Guetta and Georges Maltsiniotis. Homologie polygraphique des systèmes locaux. *Advances in Mathematics*, 448:109719, 2024. doi: 10.1016/j.aim.2024.109719. URL <https://arxiv.org/pdf/2309.10466v2>. Version consulted: arXiv:2309.10466v2, 29 August 2024.
 - [12] M. Hébert, J. Adámek, and J. Rosický. More on orthogonality in locally presentable categories. *Cahiers de Topologie et Géométrie Différentielle Catégoriques*, 42(1):51–80, 2001. URL https://www.numdam.org/item/CTGDC_2001__42_1_51_0/.
 - [13] OpenAI. The Grothendieck homotopy hypothesis via elementary expansions. OpenAI Math Release preprint [OAI:The-Grothendieck-homotopy-hypothesis-via-elementary-expansions-September-24-2026](https://arxiv.org/abs/2609.10466v2), 2026.
 - [14] Richard Steiner. Omega-categories and chain complexes. *Homology, Homotopy and Applications*, 6(1):175–200, 2004. URL <https://arxiv.org/abs/math/0403237v2>. Version consulted: arXiv:math/0403237v2, 17 May 2004.
 - [15] Richard Steiner. Orientals. In *Categories in Algebra, Geometry and Mathematical Physics*, volume 431 of *Contemporary Mathematics*, pages 427–439. American Mathematical Society, 2007. URL <https://arxiv.org/pdf/math/0601383v2>. Numbering follows arXiv:math/0601383v2, 2 July 2007.
 - [16] Ross Street. The algebra of oriented simplexes. *Journal of Pure and Applied Algebra*, 49(3):283–335, 1987. doi: 10.1016/0022-4049(87)90137-X. URL [https://doi.org/10.1016/0022-4049\(87\)90137-X](https://doi.org/10.1016/0022-4049(87)90137-X).

- [17] R. W. Thomason. Cat as a closed model category. *Cahiers de topologie et géométrie différentielle*, 21(3):305–324, 1980. URL https://www.numdam.org/item/CTGDC_1980__21_3_305_0/.
- [18] Krzysztof Worytkiewicz, Kathryn Hess, Paul-Eugène Parent, and Andrew Tonks. A model structure à la Thomason on 2-Cat. *Journal of Pure and Applied Algebra*, 208(1):205–236, 2007. doi: 10.1016/j.jpaa.2005.12.010. URL <https://doi.org/10.1016/j.jpaa.2005.12.010>.
- [19] Krzysztof Worytkiewicz, Kathryn Hess, Paul-Eugène Parent, and Andrew Tonks. Corrigendum to: “a model structure à la Thomason on 2-Cat”. *Journal of Pure and Applied Algebra*, 220(12): 4017–4023, 2016. doi: 10.1016/j.jpaa.2016.04.002. URL <https://doi.org/10.1016/j.jpaa.2016.04.002>.